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Yang et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **MAGNETIC RANDOM ACCESS MEMORY AND MANUFACTURING METHOD THEREOF**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 506 days.

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(22) Filed: **Sep. 29, 2021**

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(51) **Int. Cl.**

G11C 11/16	(2006.01)
H10B 61/00	(2023.01)
H10N 50/01	(2023.01)
H10N 50/10	(2023.01)

(Continued)

(52) **U.S. Cl.**

CPC ***H10N 50/01*** (2023.02); ***G11C 11/161*** (2013.01); ***H10B 61/00*** (2023.02); ***H10B 61/10*** (2023.02); ***H10B 61/22*** (2023.02); ***H10N 50/10*** (2023.02); ***H10N 50/80*** (2023.02); ***H10N 50/85*** (2023.02)

(58) **Field of Classification Search**

CPC H10N 50/01; H10N 50/10; H10N 50/80; H10N 50/85; H10B 61/00; H10B 61/10

See application file for complete search history.

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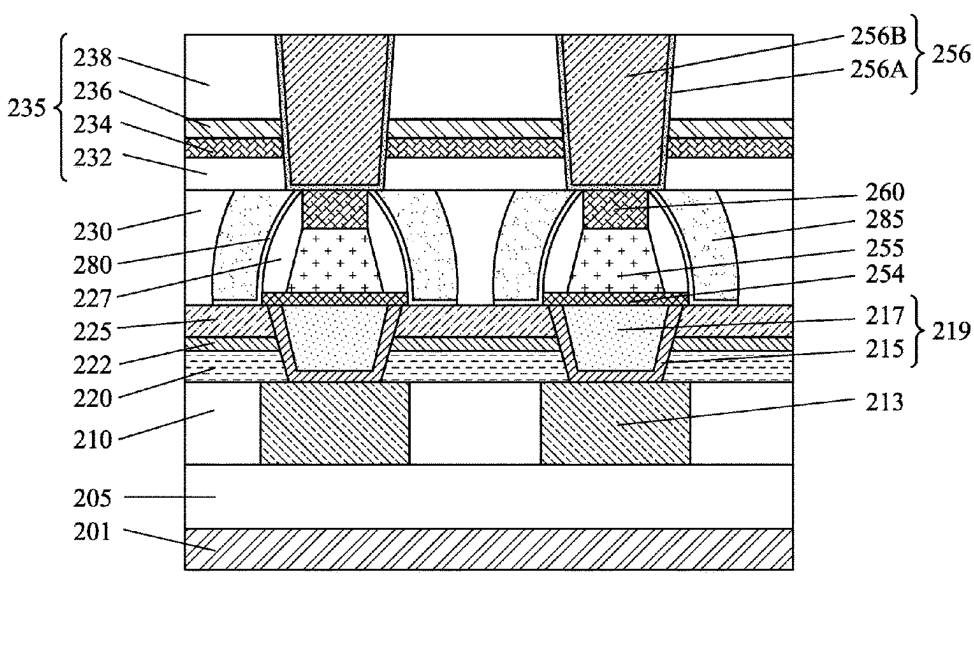
(57) **ABSTRACT**

In a method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, a first layer made of a conductive material is formed over a substrate. A second layer for a magnetic tunnel junction (MTJ) stack is formed over the first conductive layer. A third layer is formed over the second layer. A first hard mask pattern is formed by patterning the third layer. The MTJ stack is formed by patterning the second layer by an etching operation using the first hard mask pattern as an etching mask. The etching operation stops at the first layer. A sidewall insulating layer is formed over the MTJ stack. After the sidewall insulating layer is formed, a bottom electrode is formed by patterning the first layer to form the MRAM cell including the bottom electrode, the MTj stack and the first hard mask pattern as an upper electrode.

20 Claims, 26 Drawing Sheets



(51) **Int. Cl.**
H10N 50/80 (2023.01)
H10N 50/85 (2023.01)

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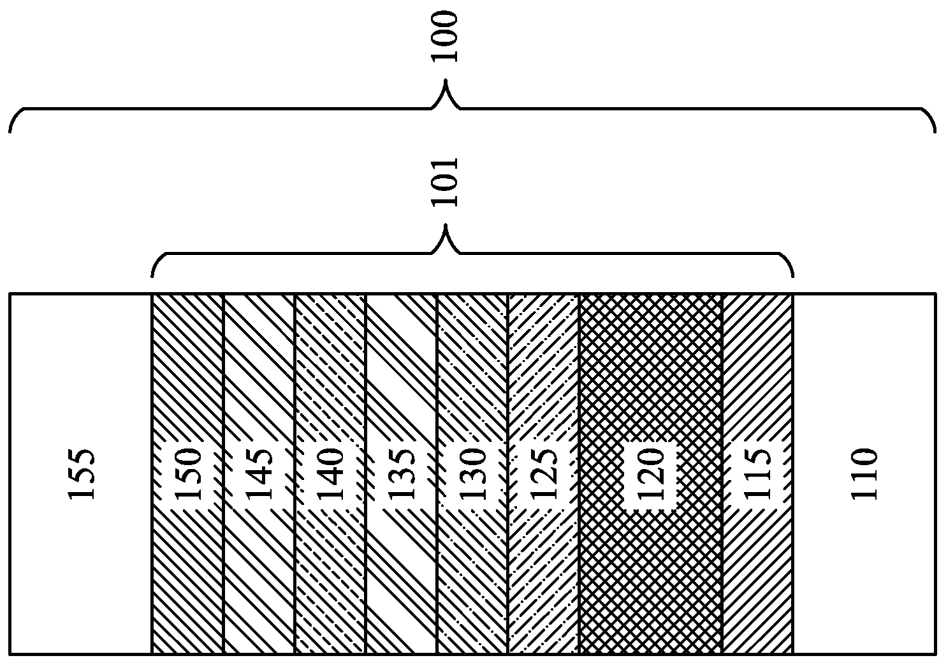
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FIG. 1B

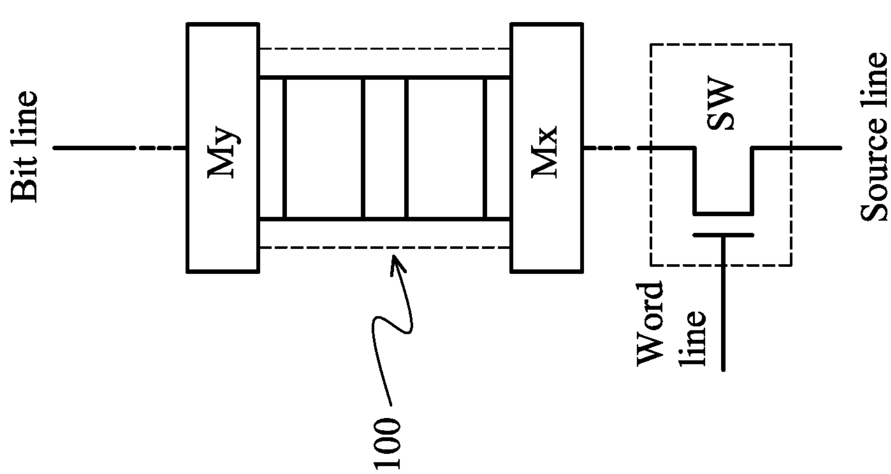
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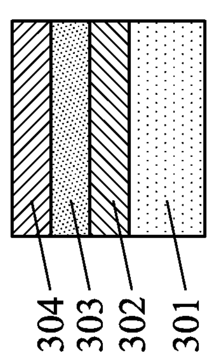
FIG. 1A



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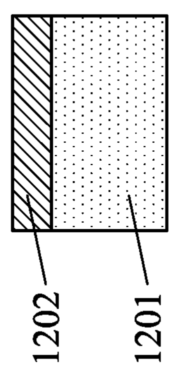

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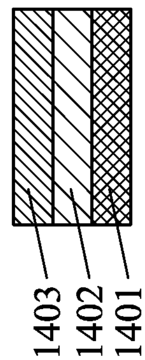
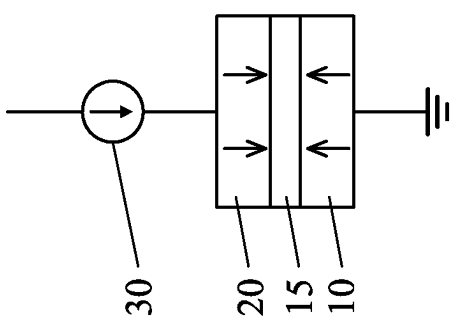


FIG. 2C



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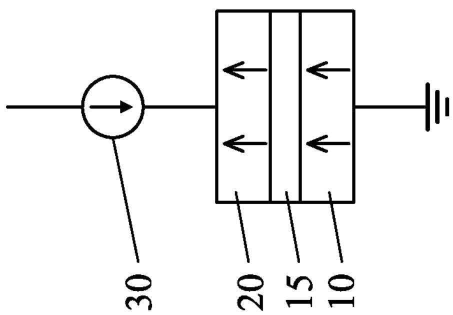

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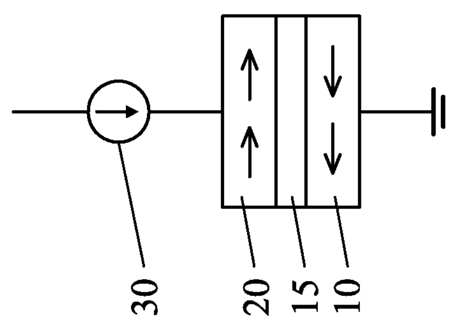
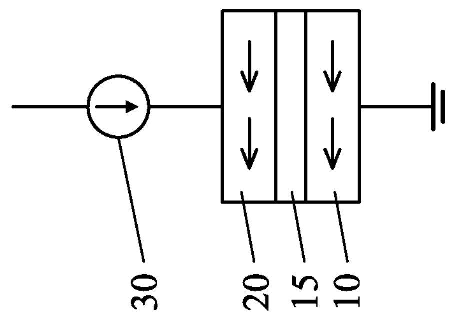


FIG. 3C



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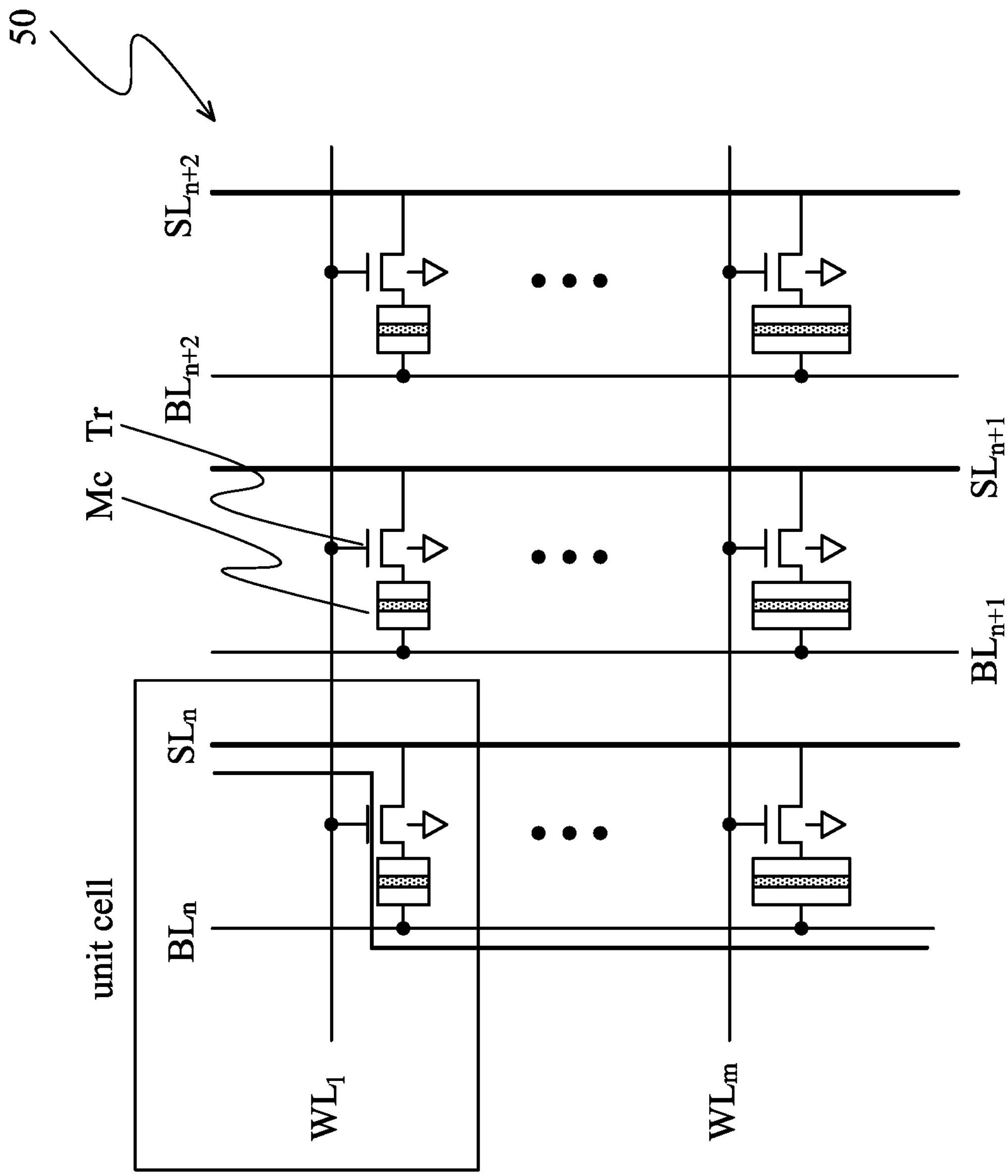

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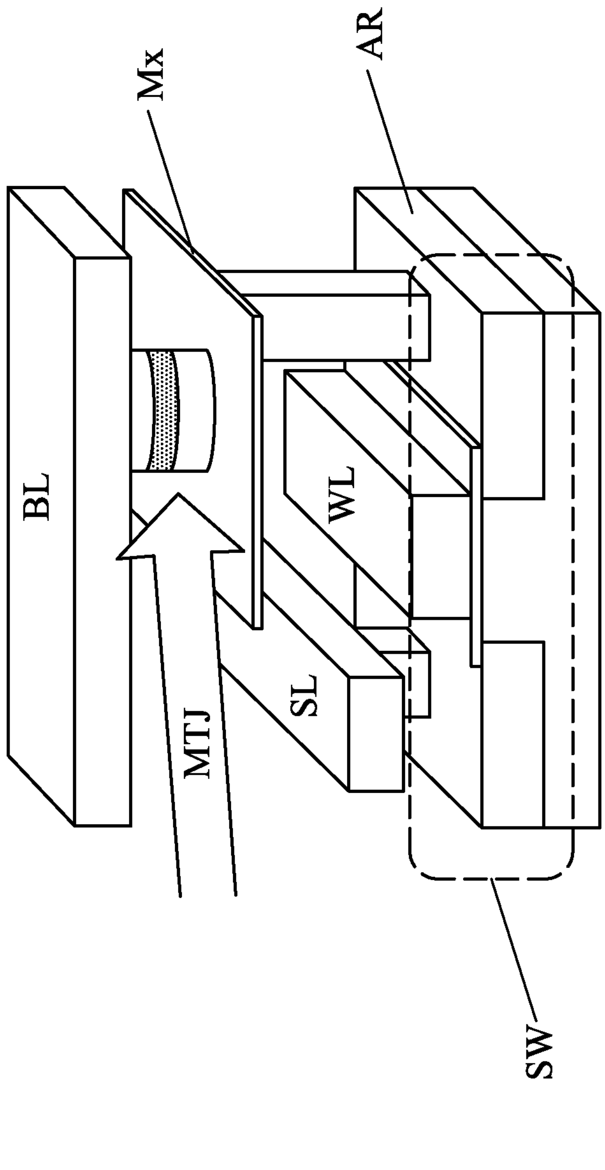

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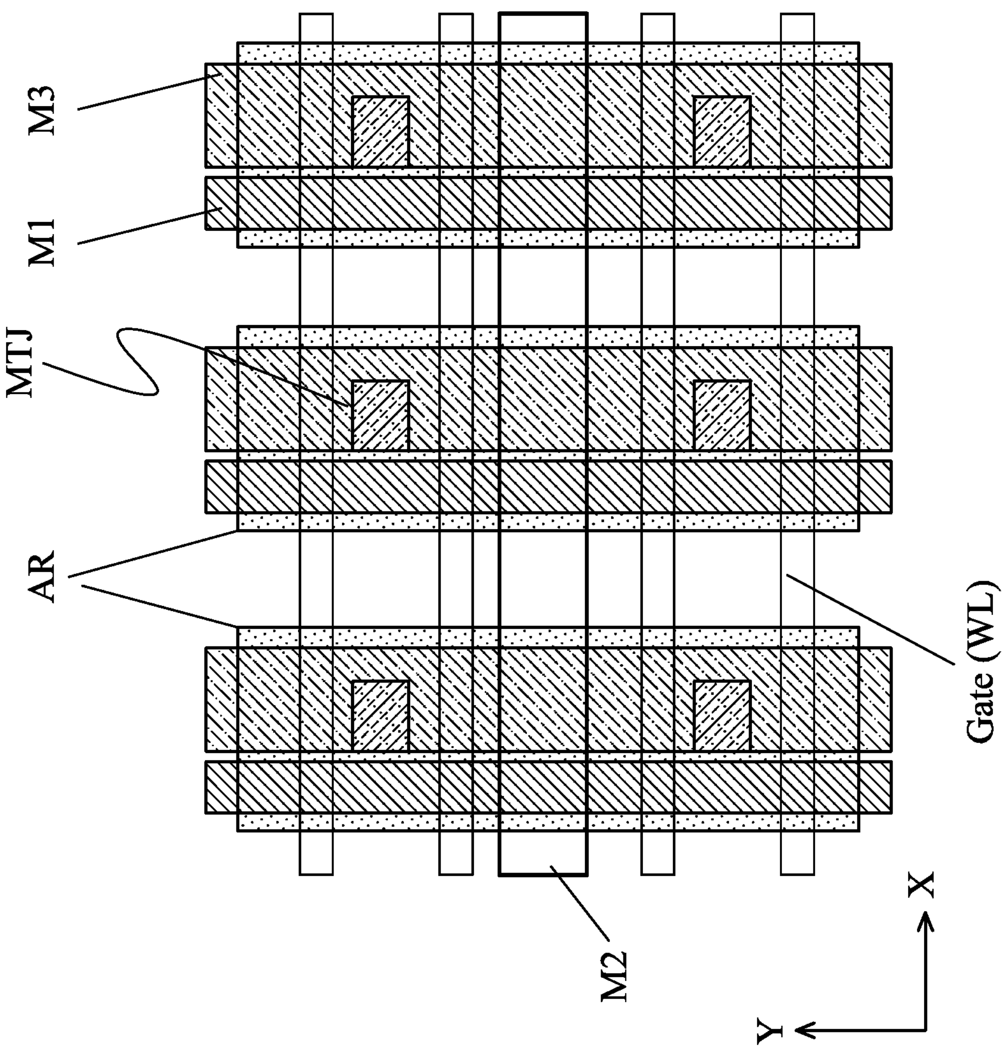

FIG. 4C

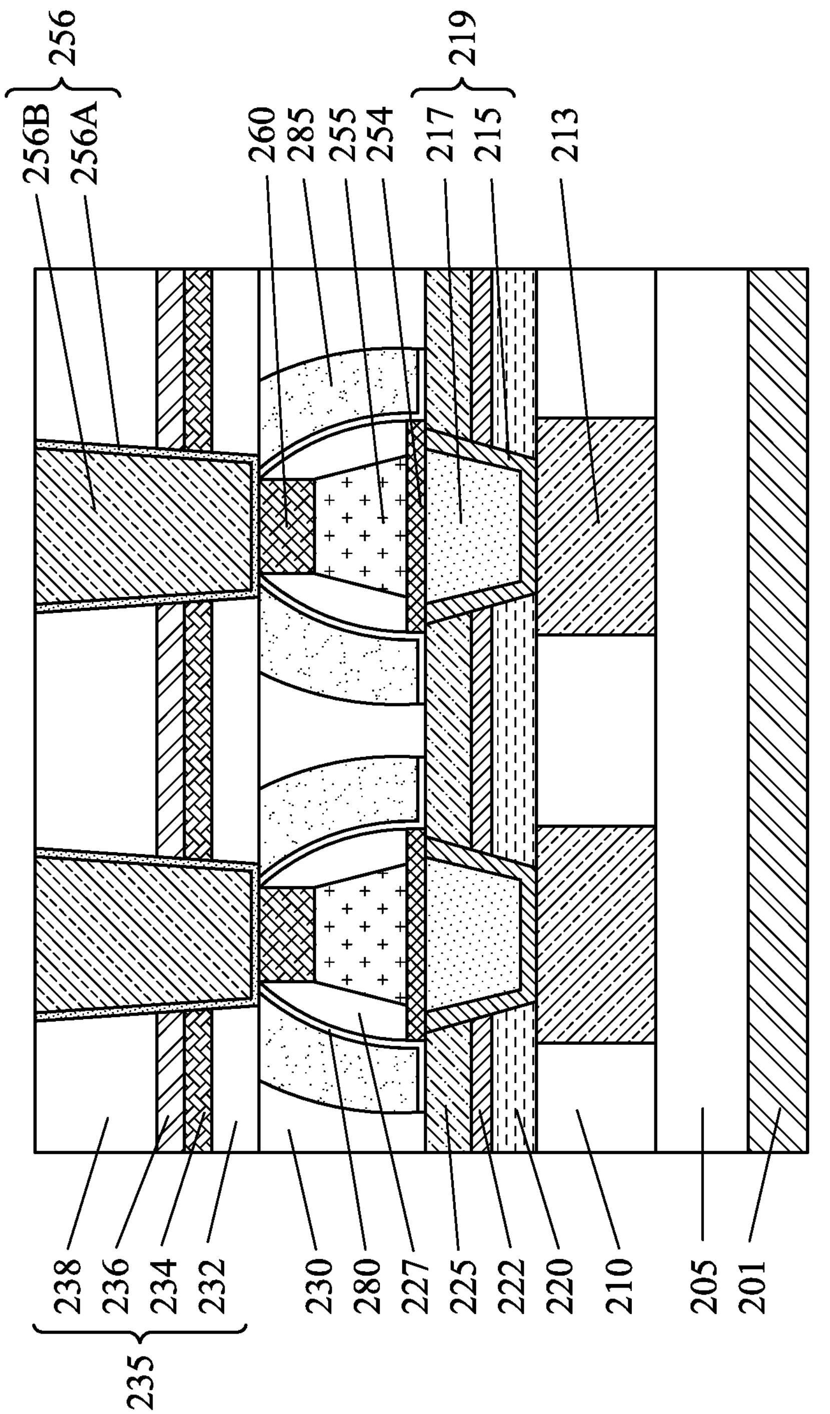
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FIG. 5A

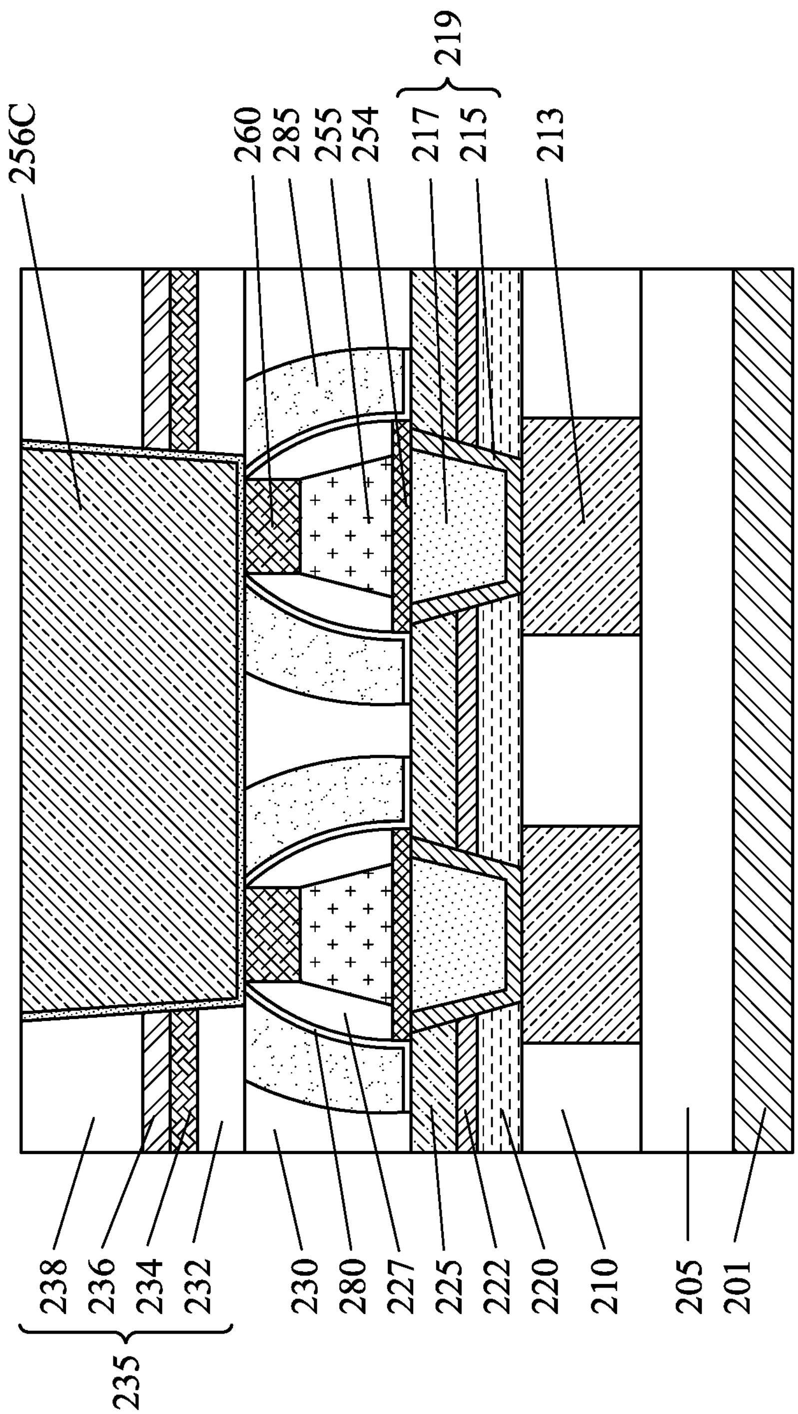
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FIG. 5B

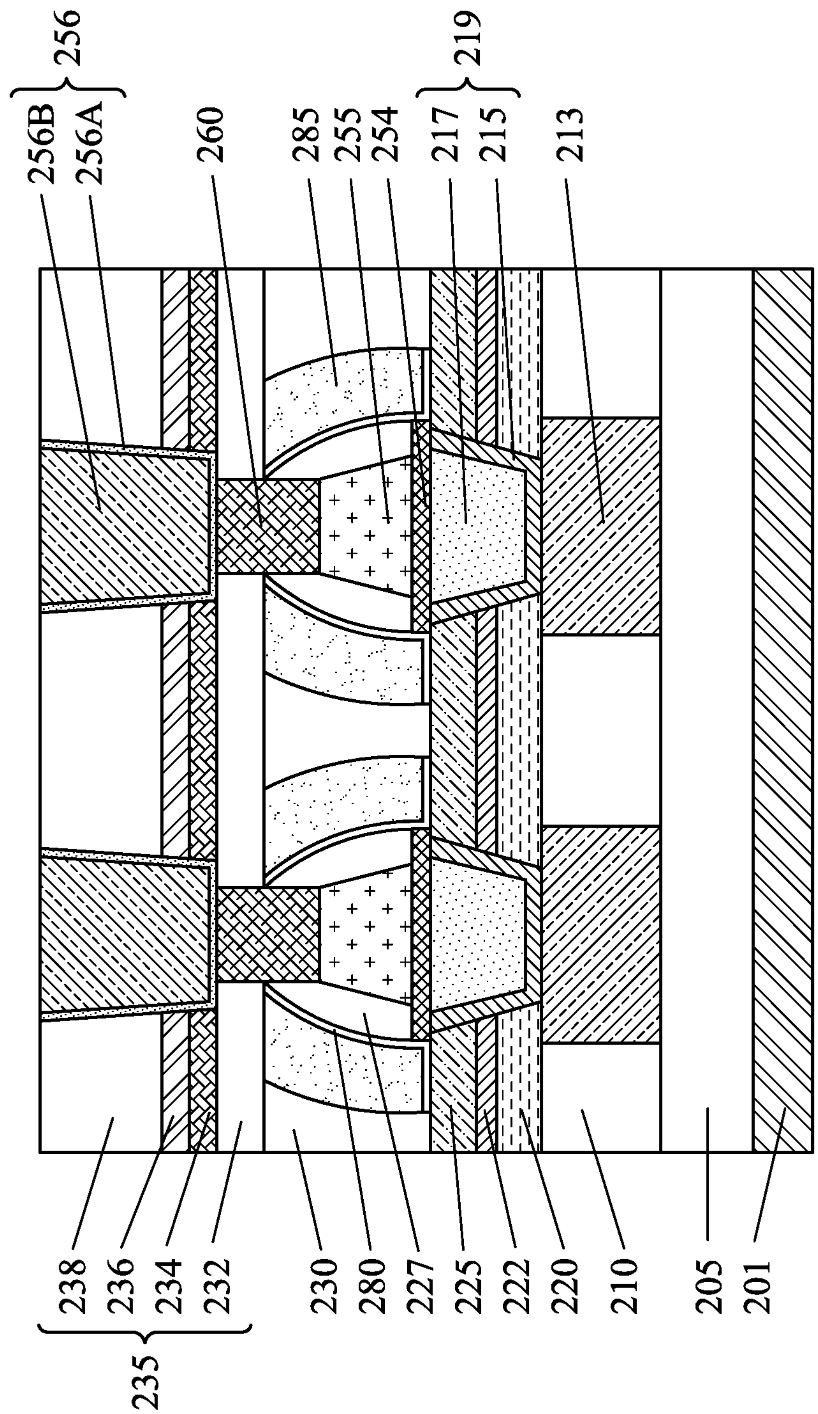
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FIG. 5C

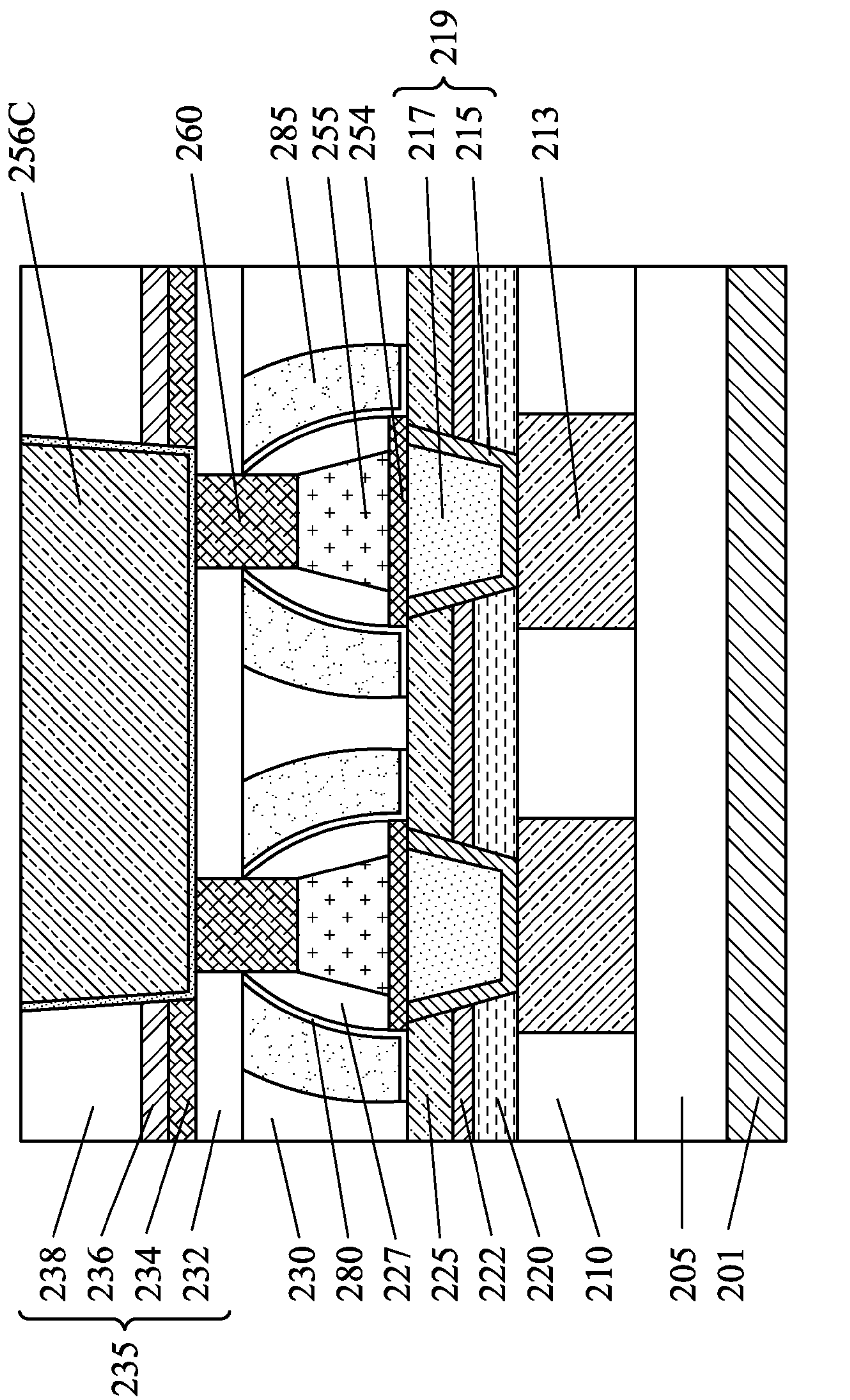
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FIG. 5D

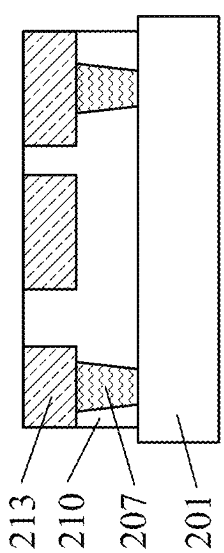
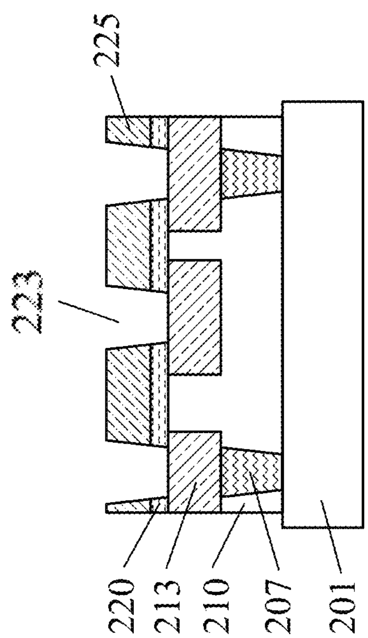


FIG. 6A



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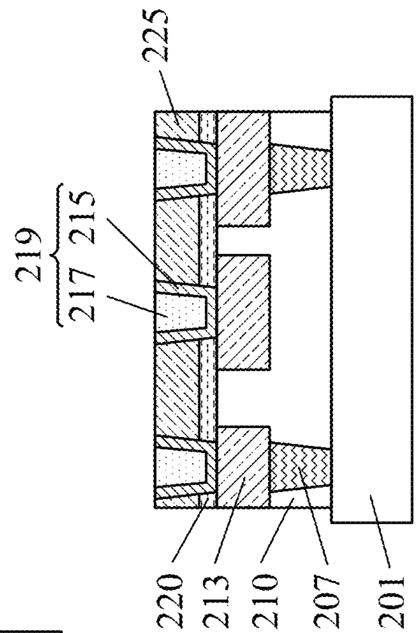

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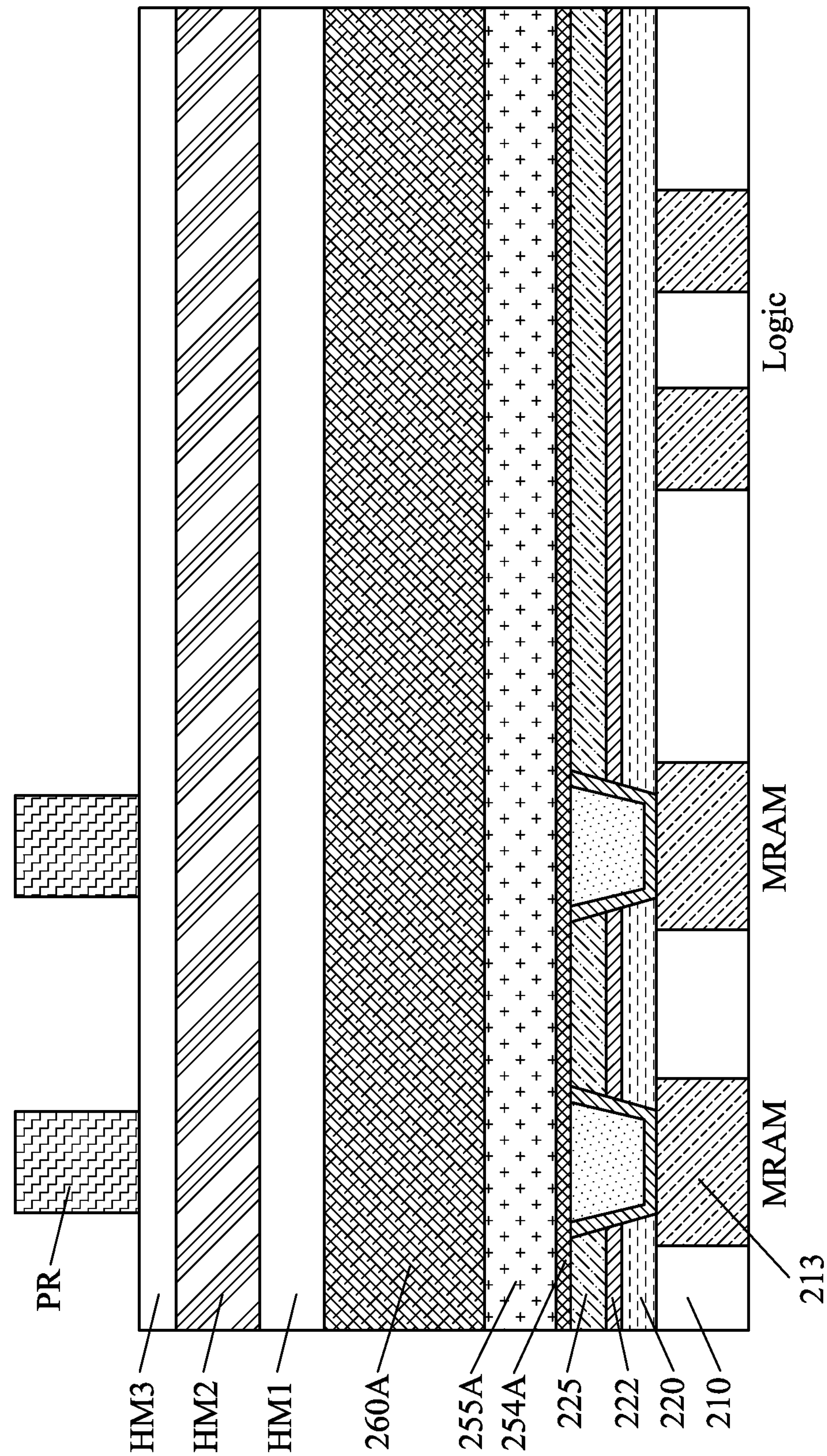
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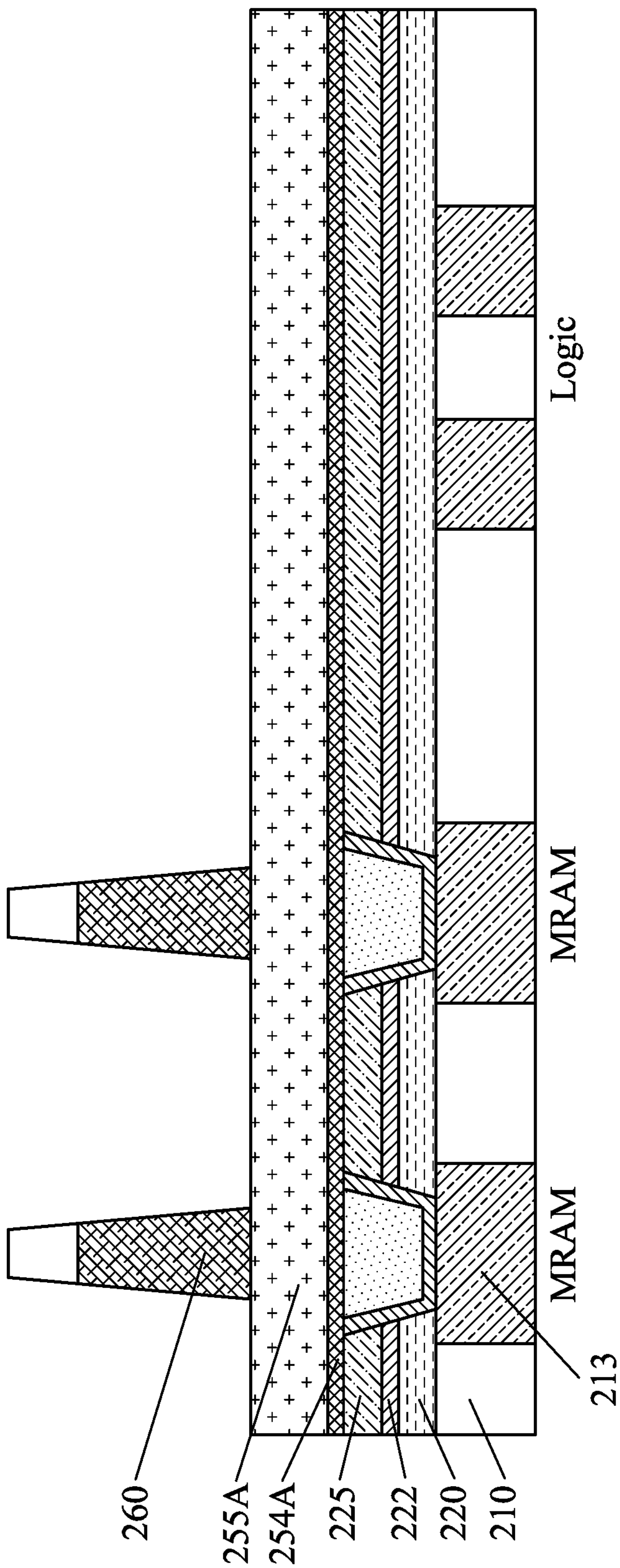

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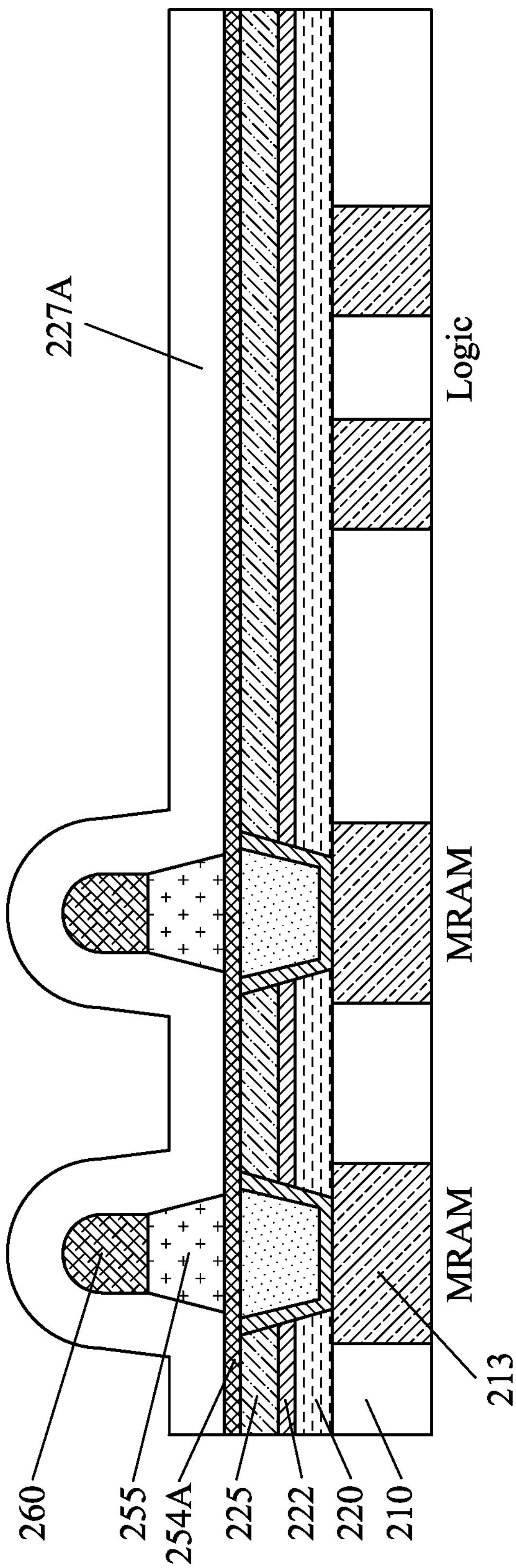

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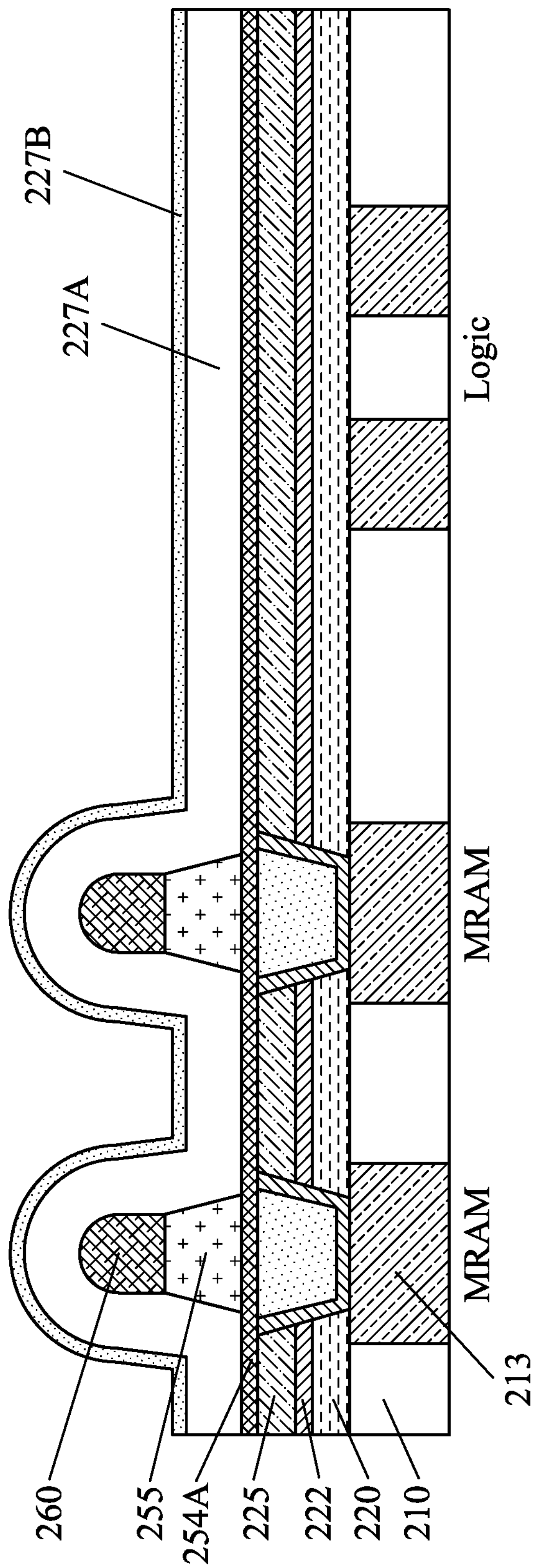


FIG. 10

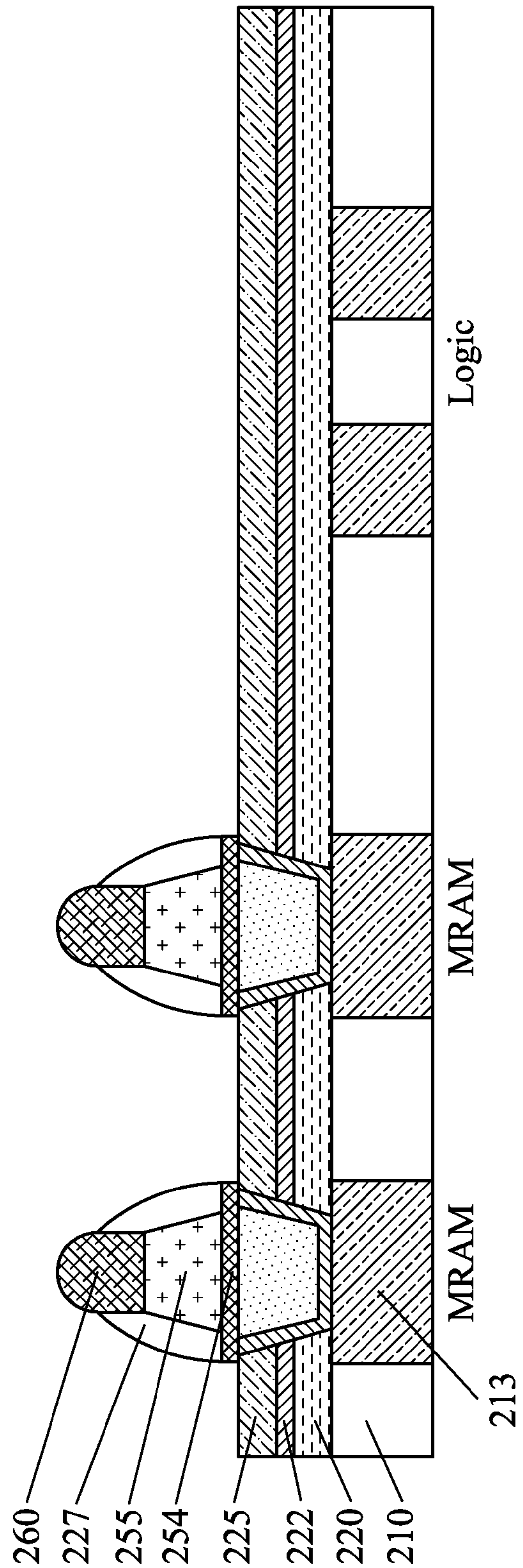


FIG. 11

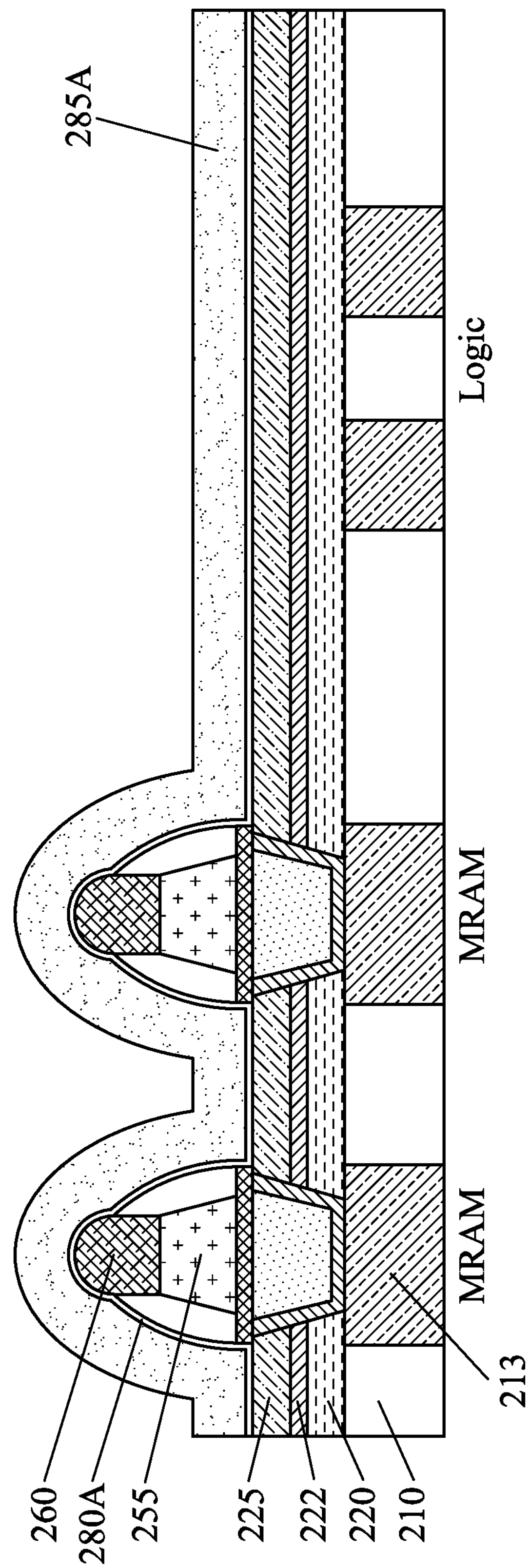
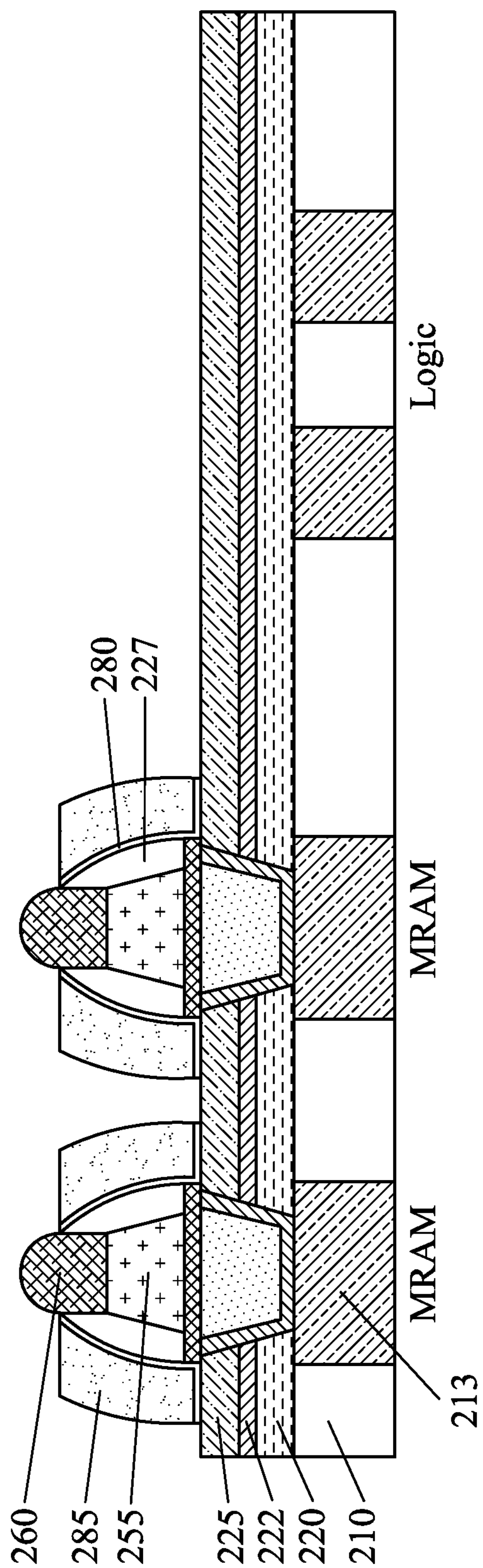


FIG. 12



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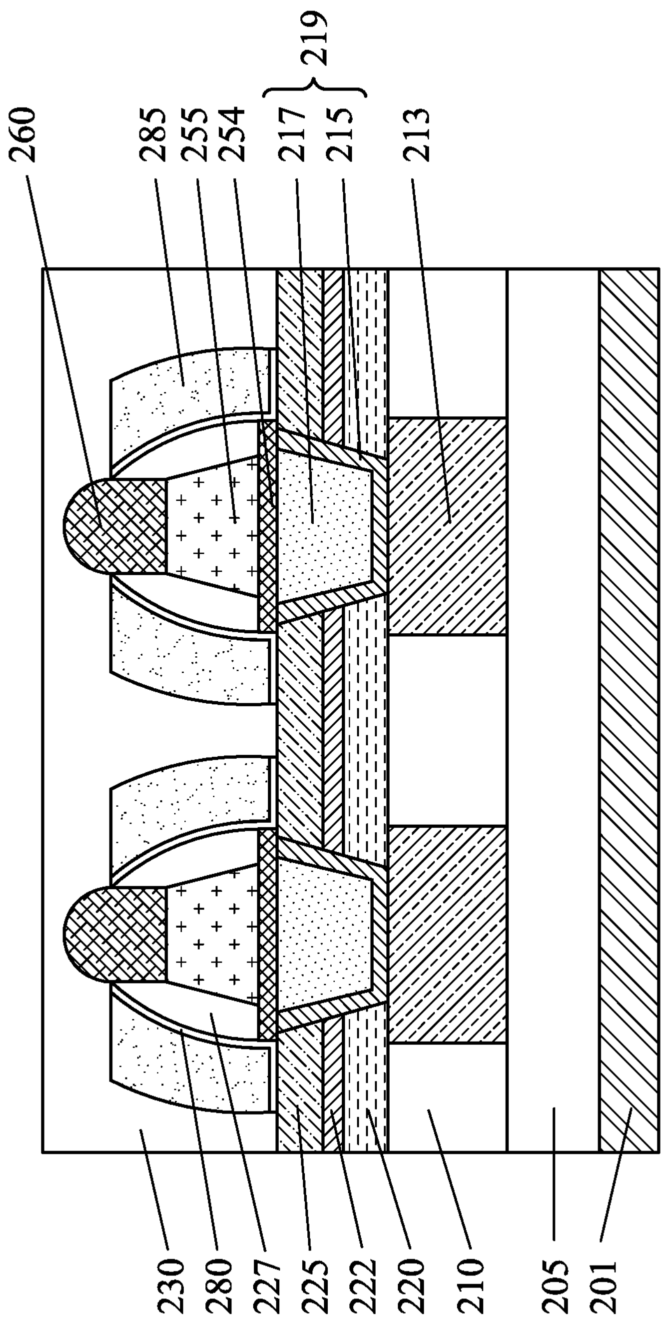
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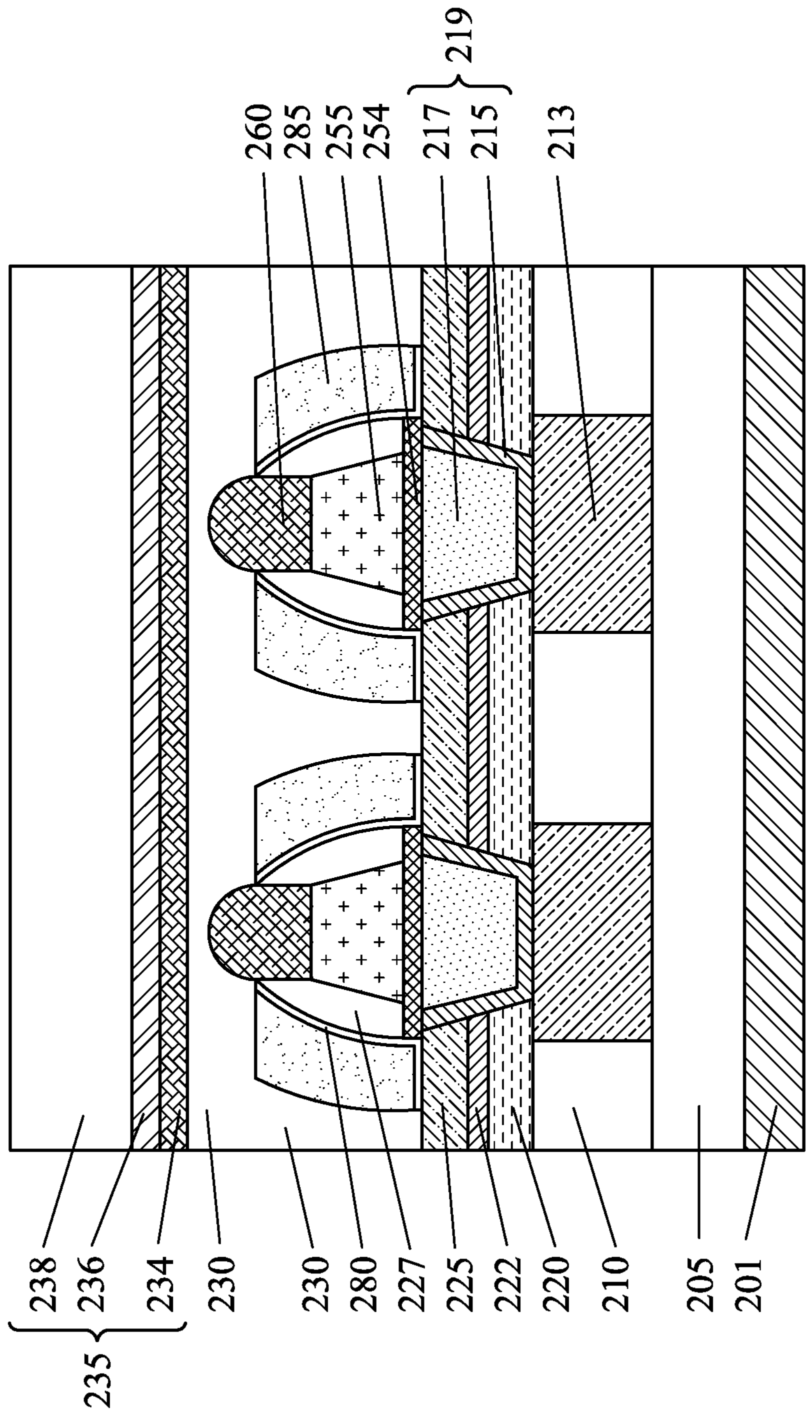
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FIG. 15

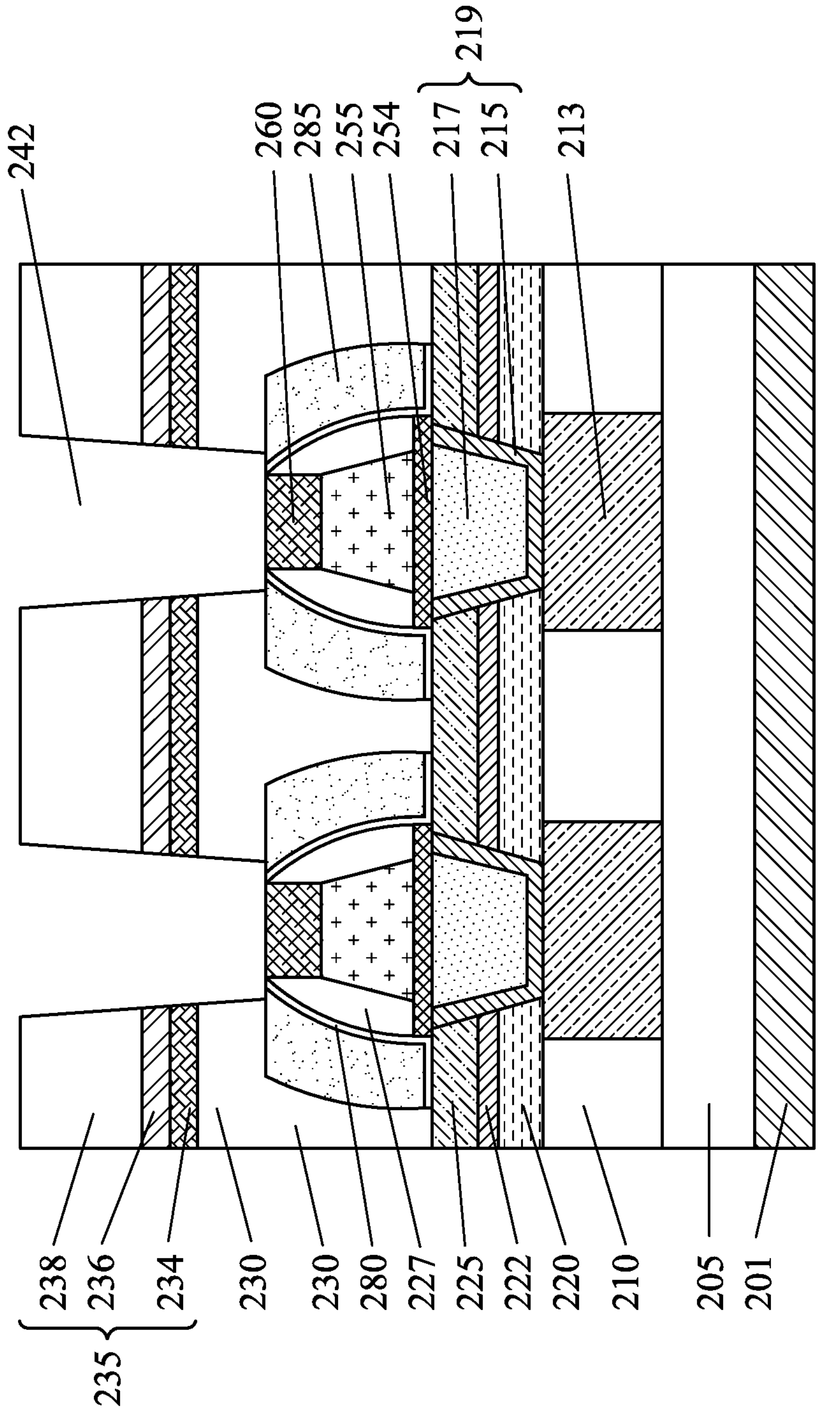
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FIG. 16

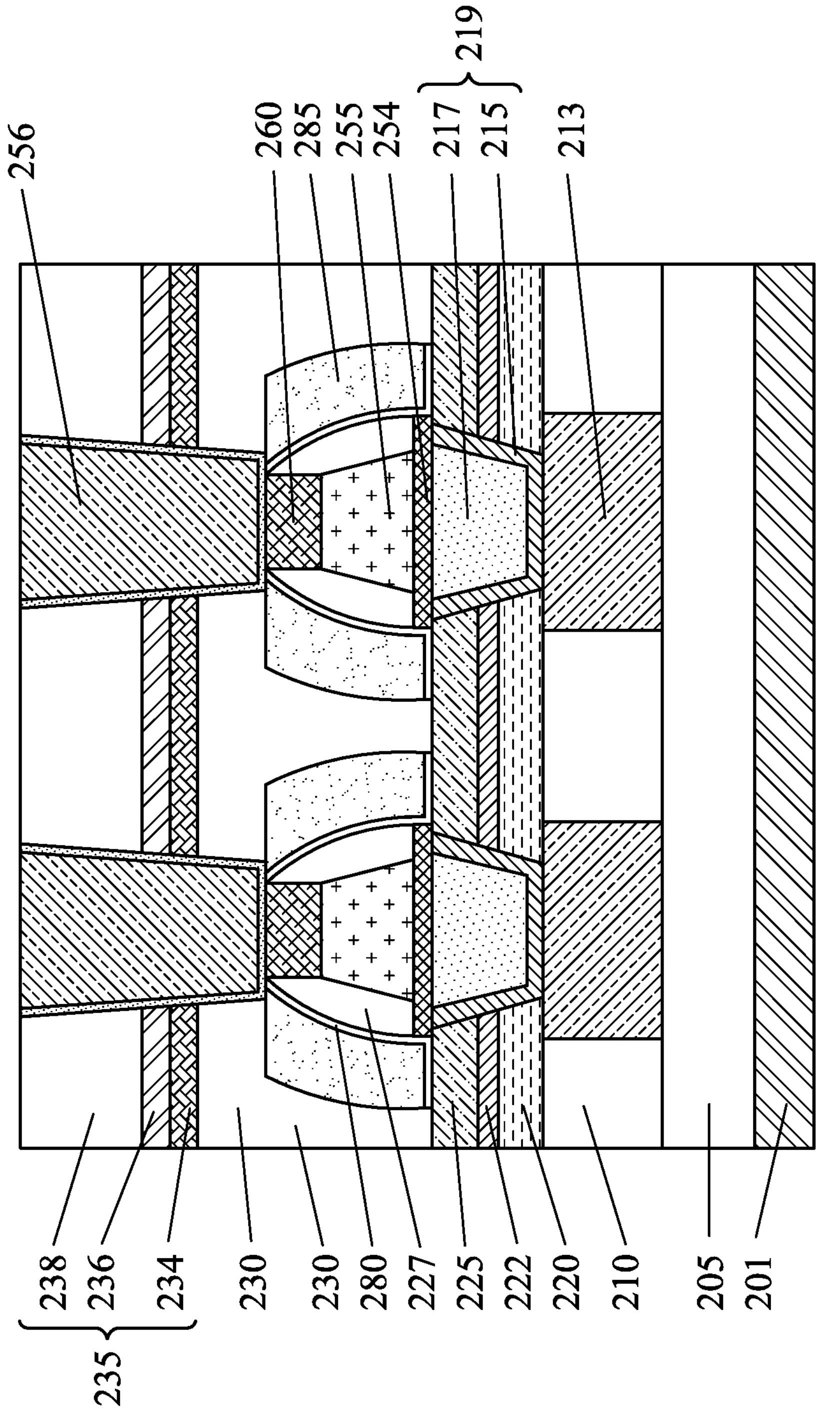

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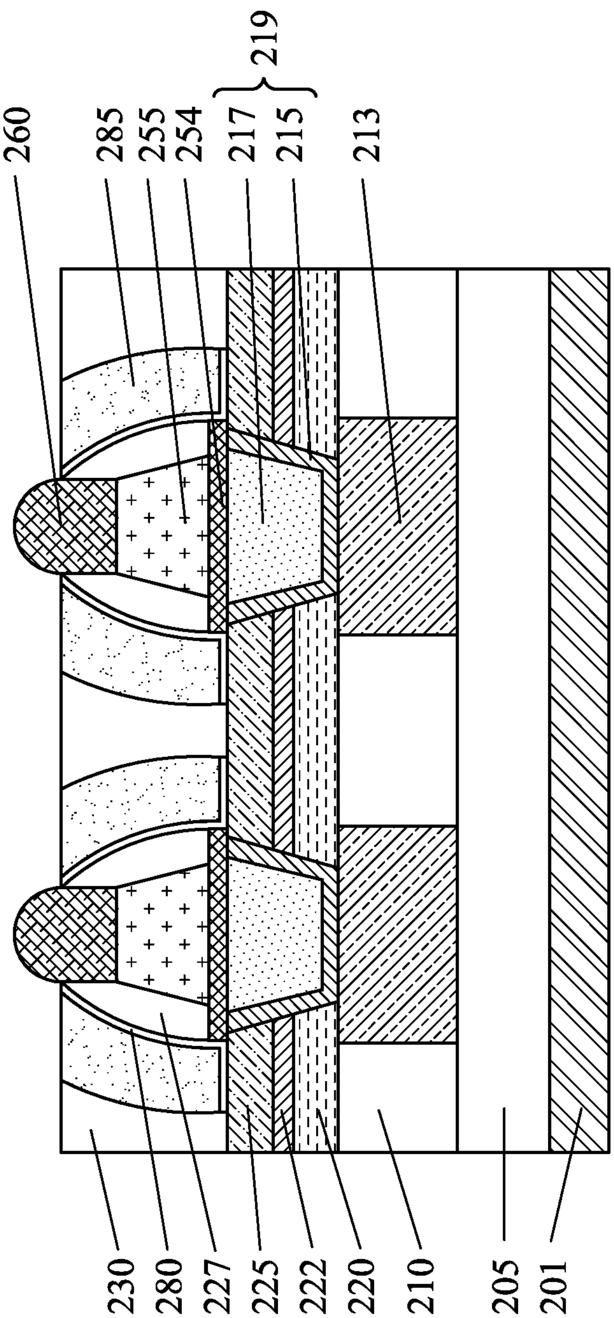
FIG. 17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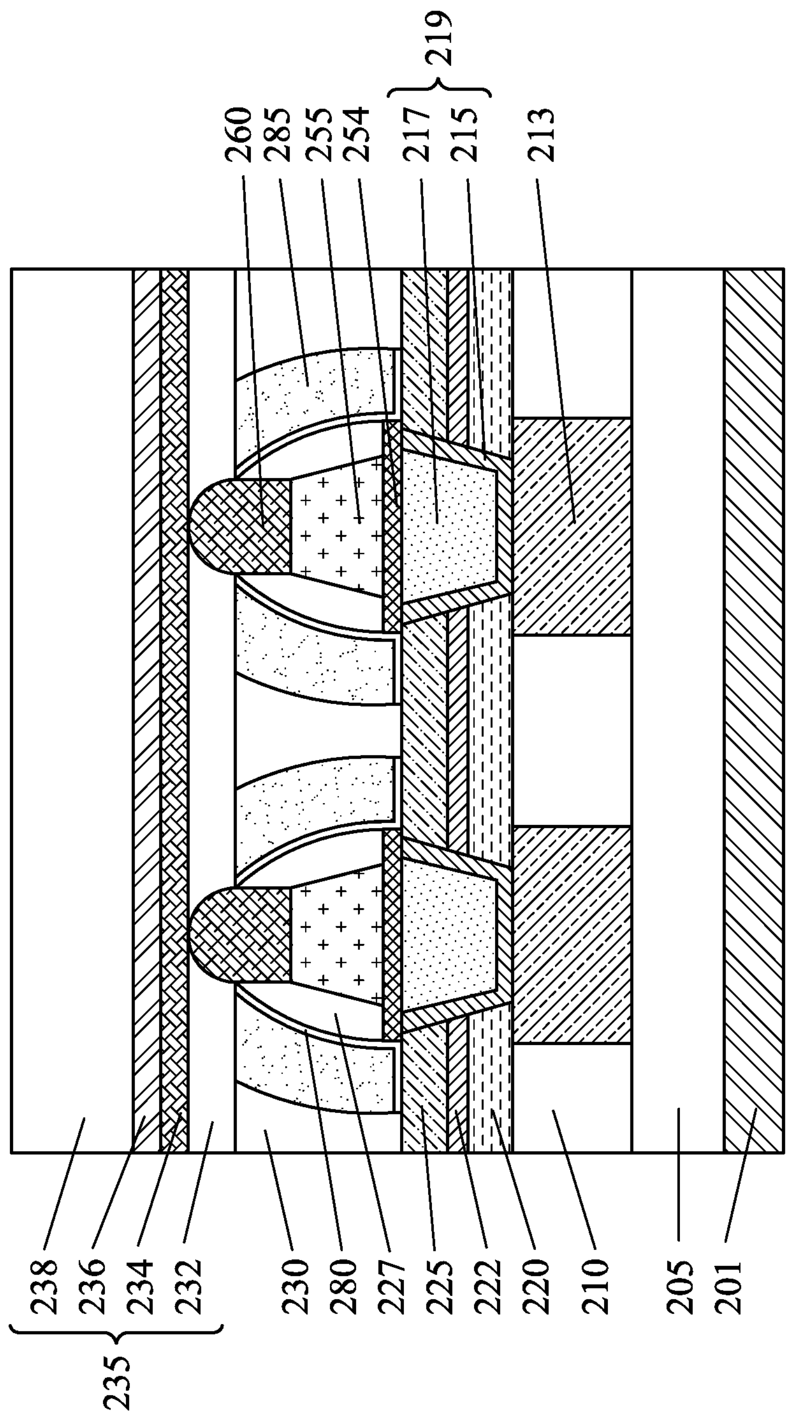
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FIG. 19

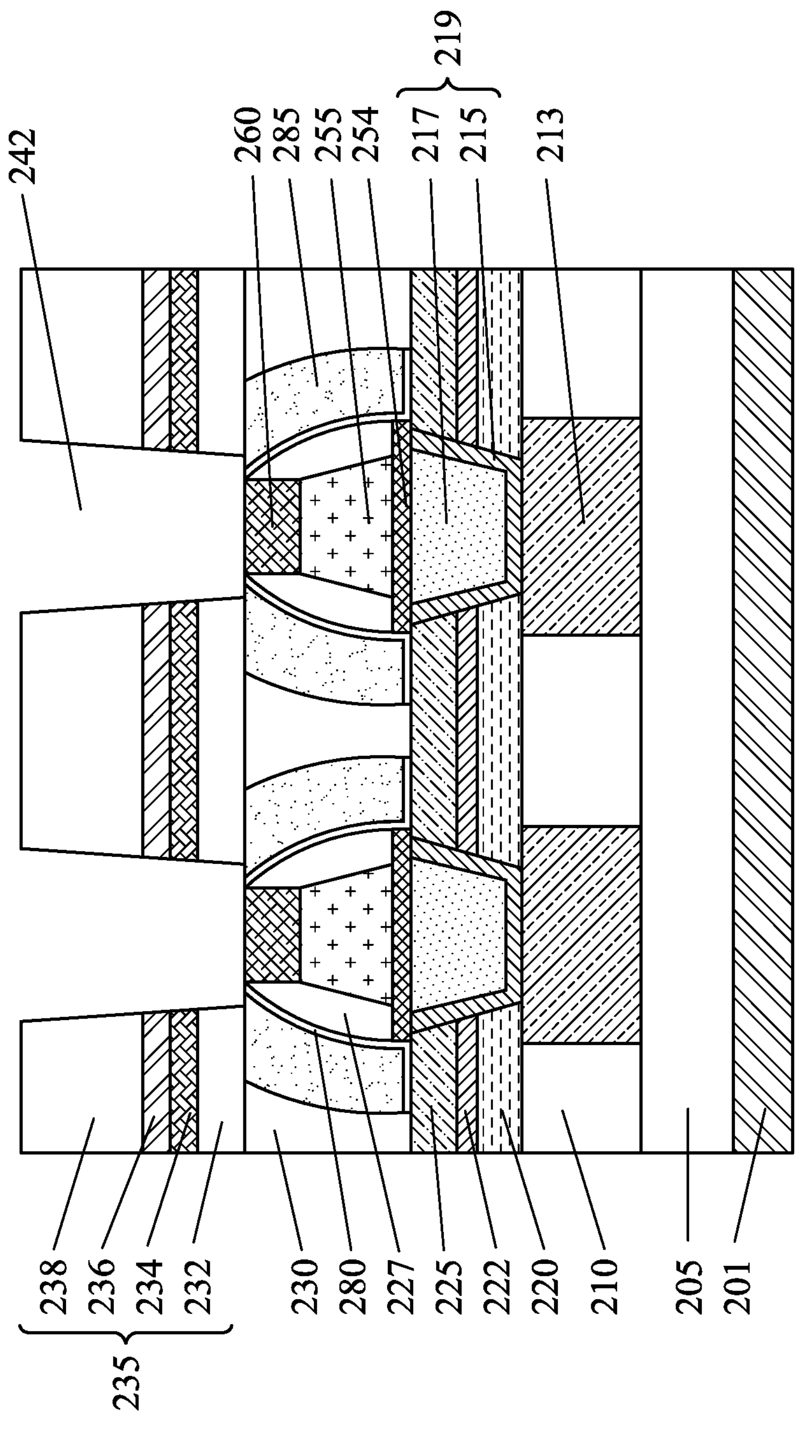
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FIG. 20

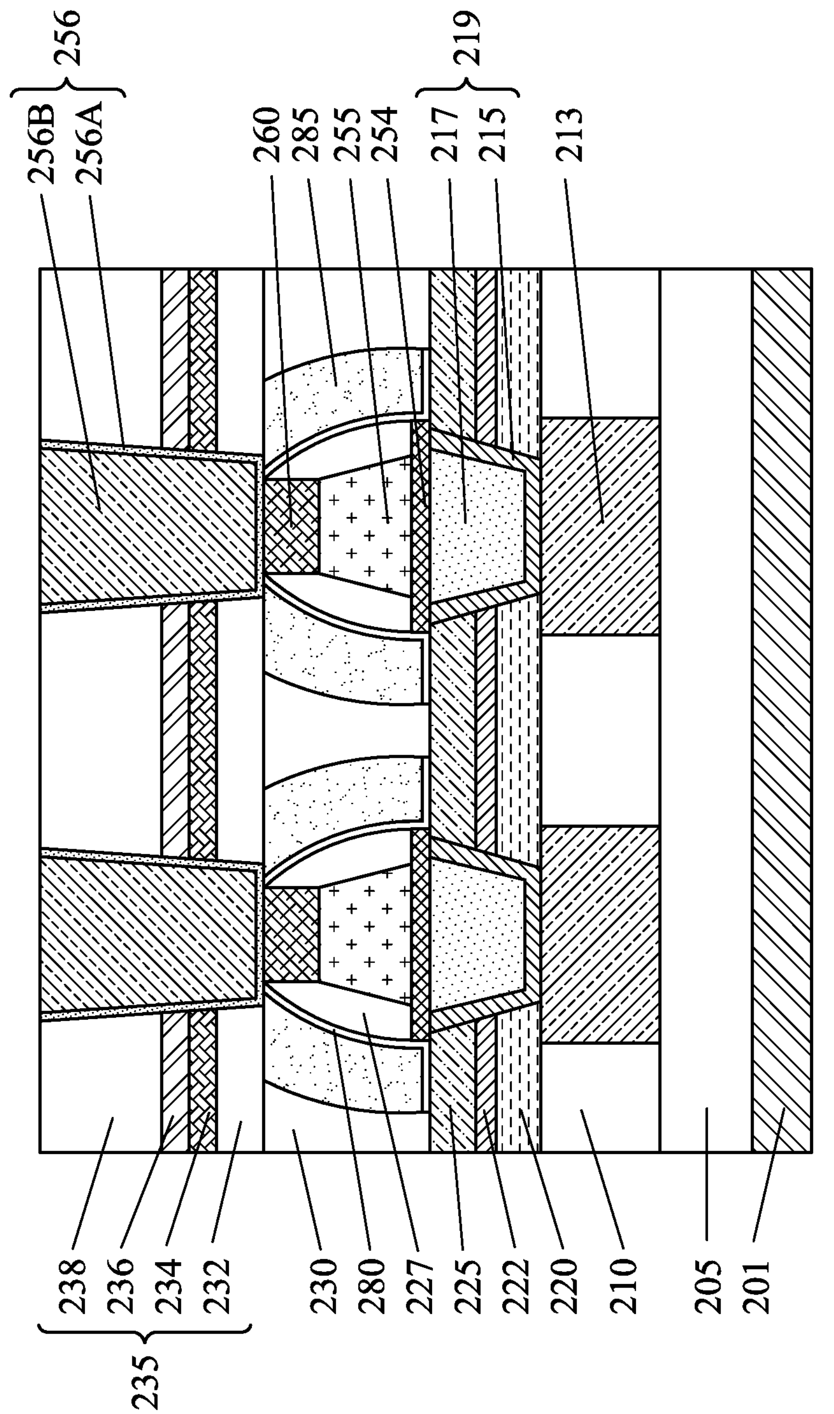
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FIG. 21

MAGNETIC RANDOM ACCESS MEMORY AND MANUFACTURING METHOD THEREOF

RELATED APPLICATION

This application claims priority to U.S. Provisional Patent Application No. 63/172,810 filed on Apr. 9, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

A magnetic random access memory (MRAM) is a device based on a magnetic tunnel junction cell formed with a semiconductor device, and offers comparable performance to volatile static random access memory (SRAM) and comparable density with lower power consumption to volatile dynamic random access memory (DRAM). Compared to non-volatile memory (NVM) flash memory, an MRAM offers much faster access times and suffers minimal degradation over time, whereas a flash memory can only be rewritten a limited number of times. An MRAM cell is formed by a magnetic tunneling junction (MTJ) comprising two ferromagnetic layers which are separated by a thin insulating barrier, and operates by tunneling of electrons between the two ferromagnetic layers through the insulating barrier.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a schematic view of an MTJ MRAM cell according to an embodiment of the present disclosure.

FIG. **1**B is a schematic cross sectional view of the MTJ film stack according to an embodiment of the present disclosure.

FIGS. **2**A, **2**B and **2**C show schematic cross sectional views of magnetic layers of the MTJ film stack according to an embodiment of the present disclosure.

FIGS. **3**A and **3**B show operations of the MTJ film stack.

FIGS. **3**C and **3**D show operations of the MTJ film stack.

FIG. **4**A shows a schematic circuit diagram of an MTJ MRAM, FIG. **4**B shows a schematic perspective view of a memory cell of the MTJ MRAM and FIG. **4**C shows a memory cell layout of the MTJ MRAM.

FIGS. **5**A, **5**B, **5**C and **5**D show cross sectional views of a semiconductor device including an MRAM according to embodiments of the present disclosure.

FIGS. **6**A, **6**B and **6**C show various stages of a sequential manufacturing process of a semiconductor device including an MRAM according to an embodiment of the present disclosure.

FIGS. **7**, **8**, **9**, **10**, **11**, **12**, **13**, **14**, **15**, **16**, and **17** show various stages of a sequential manufacturing process of a semiconductor device including an MRAM according to an embodiment of the present disclosure.

FIGS. **18**, **19**, **20** and **21** show various stages of a sequential manufacturing process of a semiconductor device including an MRAM according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity. In the accompanying drawings, some layers/features may be omitted for simplification.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term "made of" may mean either "comprising" or "consisting of" Further, in the following fabrication process, there may be one or more additional operations in/between the described operations, and the order of operations may be changed. In the present disclosure, a phrase "one of A, B and C" means "A, B and/or C" (A, B, C; A and B; A and C; B and C, or A, B and C), and does not mean one element from A, one element from B and one element from C, unless otherwise described.

FIG. **1**A is a schematic view of a MTJ MRAM cell according to an embodiment of the present disclosure, and FIG. **1**B is a schematic cross sectional view of the MTJ film stack. The MTJ cell **100** is disposed between a lower metal layer Mx and an upper metal layer My of a semiconductor device. The metal layers Mx and My are used to connect one element to another element in a semiconductor device formed at a different level above a substrate. Further, the lower metal layer Mx is coupled to a switching device SW, which can be formed by a MOS FET including, but not limited to, a planar MOS FET, a fin FET, a gate-all-around (GAA) FET, or any other switching devices. A control terminal (e.g., a gate terminal of FET) of the switching device is coupled to a word line. The upper metal layer My is coupled to a bit line. In some embodiments, the switching device SW is disposed between the upper metal layer My and the bit line. In some embodiments, the upper metal layer My is the bit line.

The MTJ cell **100** shown in FIG. **1**B includes a first electrode layer **110** coupled to the lower metal layer Mx and a second electrode layer **155** coupled to the upper metal layer My. An MTJ film stack **101** is disposed between the first electrode layer **110** and the second electrode layer **155**.

The MTJ film stack (MTJ functional layer) **101** includes a first pinned magnetic layer **130**, a free magnetic layer **140**, and a tunneling barrier layer **135** made of a non-magnetic material and disposed between the first pinned magnetic layer **130** and the free magnetic layer **140**. The free magnetic layer **140** and the first pinned magnetic layer **130** include one or more ferromagnetic materials that can be magnetically oriented, respectively. The first pinned magnetic layer **130** is configured such that the magnetic orientation is fixed and will not respond to a typical magnetic field. In some embodiments, the thickness of the free magnetic layer **140** is in a range from about 0.8 nm to about 1.5 nm. In some embodiments, the thickness of the first pinned magnetic layer **130** is in a range from about 0.8 nm to about 2.0 nm.

The tunneling barrier layer **135** includes a relatively thin oxide layer capable of electrically isolating the free magnetic layer **140** from the first pinned magnetic layer **130** at low potentials and capable of conducting current through electron tunneling at higher potentials. In some embodiments, the tunneling barrier layer **135** includes magnesium oxide (MgO) having a thickness in a range from about 0.5 nm to about 1.2 nm.

The MTJ film stack **101** further includes an antiferromagnetic layer **125**, as shown in FIG. **1**B. The anti-ferromagnetic layer **125** is used to fix the magnetic orientation of the first pinned magnetic layer **130**. The antiferromagnetic layer **125** includes ruthenium (Ru) or any other suitable antiferromagnetic material. In some embodiments, the thickness of the antiferromagnetic layer **125** is in a range from about 0.4 nm to about 1.0 nm.

The MTJ film stack **101** further includes a second pinned magnetic layer **120** including one or more magnetic materials, as shown in FIG. **1**B.

The first electrode layer **110** is formed on the lower metal layer Mx made of, for example, Cu, Al, W, Co, Ni, and/or an alloy thereof; and the upper metal layer My made of, for example, Cu, Al, W, Co, Ni, and/or an alloy thereof, is formed on the second electrode layer **155**.

The first pinned magnetic layer **130** includes multiple layers of magnetic materials. In some embodiments, as shown in FIG. **2**A, the first pinned magnetic layer **130** includes four layers **1301**, **1302**, **1303** and **1304**, where the uppermost layer **1304** is in contact with the tunneling barrier layer **135** and the bottommost layer **1301** is in contact with the antiferromagnetic layer **125**. In some embodiments, the bottommost layer **1301** includes a multilayer structure of cobalt (Co) and platinum (Pt). In some embodiments, a thickness of the cobalt layer is in a range from about 0.3 nm to about 0.6 nm and a thickness of the platinum layer is in a range from about 0.2 nm to about 0.5 nm. The thickness of the cobalt layer can be the same as or greater than the platinum layer. The cobalt layers and the platinum layers are alternately stacked such that the total thickness of the bottommost layer **1301** is in a range from about 2.0 nm to about 5.0 nm in some embodiments. The layer **1302** includes a cobalt layer having a thickness in a range from about 0.4 nm to about 0.6 nm. In certain embodiments, the bottommost layer **1301** includes the cobalt layer and the layer **1302** is the multilayer of the cobalt layers and the platinum layers as set forth above. In this disclosure, an "element" layer generally means that the content of the "element" is more than 99%.

The layer **1303** is a spacer layer. The thickness of the spacer layer **1303** is in a range from about 0.2 nm to about 0.5 nm in some embodiments.

The uppermost layer **1304** includes a cobalt iron boron (CoFeB) layer, a cobalt/palladium (CoPd) layer and/or a cobalt iron (CoFe) layer. The thickness of the layer **1304** is in a range from about 0.8 nm to about 1.5 nm in some embodiments.

The second pinned magnetic layer **120** includes multiple layers of magnetic materials in some embodiments. In some embodiments, as shown in FIG. **2**B, the second pinned magnetic layer **120** includes two layers **1201** and **1202**, where the upper layer **1202** is in contact with the antiferromagnetic layer **125**. In some embodiments, the lower layer **1201** includes a multilayer structure of cobalt (Co) and platinum (Pt). In some embodiments, a thickness of the cobalt layer is in a range from about 0.3 nm to about 0.6 nm and a thickness of the platinum layer is in a range from about 0.2 nm to about 0.5 nm. The thickness of the cobalt layer can be the same as or greater than the platinum layer. The cobalt layers and the platinum layers are alternately stacked such that the total thickness of the lower layer **1201** is in a range from about 5.0 nm to about 10.0 nm in some embodiments. The upper layer **1202** includes a cobalt layer having a thickness in a range from about 0.4 nm to about 0.6 nm.

The free magnetic layer **140** includes a cobalt iron boron (CoFeB) layer, a cobalt/palladium (CoPd) layer and/or a cobalt iron (CoFe) layer having a thickness in a range from about 1.0 nm to about 2.0 nm in some embodiments. In other embodiments, the free magnetic layer **140** includes multiple layers of magnetic materials. In some embodiments, as shown in FIG. **2**C, the free magnetic layer **140** includes three layers **1401**, **1402** and **1403**, where the lower layer **1401** is in contact with the tunneling barrier layer **135**. The lower and upper layers **1401** and **1403** are a cobalt iron boron (CoFeB) layer, a cobalt/palladium (CoPd) layer and/or a cobalt iron (CoFe) layer having a thickness in a range from about 1.0 nm to about 2.0 nm in some embodiments. The middle layer **1402** is a spacer layer. The thickness of the spacer layer **1402** is in a range from about 0.2 nm to about 0.6 nm in some embodiments.

In some embodiments, the spacer layer **1303** and/or the spacer layer **1402** include an iridium layer and/or a binary alloy layer of iridium and tantalum. A spacer layer for the MTJ film stack has a super smooth surface morphology, high electric conductivity, and is substantially free from diffusion issues in some embodiments. Further, the spacer layer should also be tolerant to a low level of oxidation without significant degradation of its conductivity. The thickness of the spacer layers **1303** and/or **1402** is in a range from about 0.1 nm to about 10 nm in some embodiments, and is in a range from about 0.5 nm to about 5.0 nm in other embodiments.

The MTJ film stack **101** further includes a seed layer **115** formed on the first electrode layer **110**, a capping layer **145** formed on the free magnetic layer **140**, and a diffusion barrier layer **150** formed on the capping layer **145**, as shown in FIG. **1**B. The capping layer **145** includes a dielectric material, such as magnesium oxide or aluminum oxide, and has a thickness in a range from about 0.5 nm to about 1.5 nm in some embodiments. In some embodiments, the diffusion barrier layer **150** includes a metallic material, such as Ru, Ta, Mo or other suitable material, and has a thickness in a range from about 0.5 nm to about 1.5 nm. In some embodiments, one or both of the capping layer **145** and the diffusion barrier layer **150** are not used. In some embodiments, the seed layer **115** is made of one or more of iridium (Ir), tantalum (Ta), molybdenum (Mo), cobalt (Co), nickel (Ni), ruthenium (Ru) or platinum (Pt) or an alloy thereof.

The first electrode layer **110** includes a conductive material, such as a metal (e.g., Ta, Mo, Co, Pt, Ni), to reduce the resistance for programming. The second electrode layer **155** also includes a conductive material, such as a metal, to reduce the resistivity during reading.

The pinned magnetic layer, the free magnetic layer and the antiferromagnetic layer can be formed by physical vapor deposition (PVD), molecular beam epitaxy (MBE), pulsed laser deposition (PLD), atomic layer deposition (ALD), electron beam (e-beam) epitaxy, chemical vapor deposition (CVD), or derivative CVD processes, including low pressure CVD (LPCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), or any combinations thereof, or any other suitable film deposition method. The tunneling barrier layer and the diffusion barrier layer can also be formed by CVD, PVD or ALD or any other suitable film deposition method.

FIGS. **3**A-**3**D show a memory operation of MTJ cell. As shown in FIGS. **3**A-**3**D, the MTJ cell includes a pinned magnetic layer **10**, a tunneling barrier layer **15** and a free magnetic layer **20**. The pinned magnetic layer **10** corresponds to the first pinned magnetic layer **130** or the combination of the second pinned magnetic layer **120**, the antiferromagnetic layer **125** and the first pinned magnetic layer **130** of FIG. **1**B. The tunneling barrier layer **15** corresponds to the tunneling barrier layer **135** of FIG. **1**B and the free magnetic layer **20** corresponds to the free magnetic layer **140** of FIG. **1**B. In FIGS. **3**A-**3**D, the remaining layers are omitted. A current source **30** is coupled to the MTJ structure in series.

In FIG. **3**A, the pinned magnetic layer **10** and the free magnetic layer **20** are magnetically oriented in opposite directions. In some embodiments, the spin directions of the pinned magnetic layer **10** and the free magnetic layer **20** are parallel to the film stack direction (perpendicular to the surface of the films). In FIG. **3**B, the pinned magnetic layer **10** and the free magnetic layer **20** are magnetically oriented in the same direction. In other embodiments, the spin directions of the pinned magnetic layer **10** and the free magnetic layer **20** are perpendicular to the film stack direction (parallel with the surface of the films), as shown in FIGS. **3**C and **3**D. In FIG. **3**C, the pinned magnetic layer **10** and the free magnetic layer **20** are magnetically oriented in opposite directions, while in FIG. **3**D, the pinned magnetic layer **10** and the free magnetic layer **20** are magnetically oriented in the same direction.

If the same current value I_C is forced to flow through the MTJ cell by the current source **30**, it is found that the cell voltage V_1 in the case of FIG. **3**A (or FIG. **3**C) is larger than the cell voltage V_2 in the case of FIG. **3**B (or FIG. **3**D), because the resistance of an opposite-oriented MTJ cell shown in FIG. **3**A (or FIG. **3**C) is greater than the resistance of a same-oriented MTJ cell shown in FIG. **3**B (or FIG. **3**D). Binary logic data ("0" and "1") can be stored in a MTJ cell and retrieved based on the cell orientation and resulting resistance. Further, since the stored data does not require a storage energy source, the cell is non-volatile.

FIG. **4**A shows a schematic circuit diagram of an MTJ MRAM array **50**. Each memory cell includes a MTJ cell Mc and a transistor Tr, such as a MOS FET. The gate of the transistor Tr is coupled to one of word lines $WL_1 \ldots WL_m$ and a drain (or a source) of the transistor Tr is coupled to one end of the MTJ cell Mc, and another end of the MTJ cell is coupled to one of bit lines BL_n, BL_{n+1} and BL_{n+2}. Further, in some embodiments, signal lines (not shown) for programming are provided adjacent to the MTJ cells.

A memory cell is read by asserting the word line of that cell, forcing a reading current through the bit line of that cell, and then measuring the voltage on that bit line. For example, to read the state of a target MTJ cell, the word line is asserted to turn ON the transistor Tr. The free magnetic layer of the target MTJ cell is thereby coupled to one of the fixed potential lines SL_n, SL_{n+1} and SL_{n+2}, e.g., the ground, through the transistor Tr. Next, the reading current is forced on the bit line. Since only the given reading transistor Tr is turned ON, the reading current flows through the target MTJ cell to the ground. The voltage of the bit line is then measured to determine the state ("0" or "1") of the target MTJ cell. In some embodiments, as shown in FIG. **4**A, each MTJ cell has one reading transistor Tr. Therefore, this type of MRAM architecture is called 1T1R. In other embodiments, two transistors are assigned to one MTJ cell, forming a 2T1R system. Other cell array configurations can be employed.

FIG. **4**B shows a schematic perspective view of a memory cell of the MTJ MRAM and FIG. **4**C shows a memory cell layout of the MTJ MRAM.

As shown in FIGS. **4**B and **4**C, the MTJ cell MTJ is disposed above a switching device SW, such as a MOS FET. The gate of the MOSFET is a word line WL or coupled to a word line formed by a metal layer. The bottom electrode Mx of the MTJ cell is coupled to a drain of the MOS FET formed in an active region AR and a source of the MOS FET formed in the active region AR is coupled to the source line SL. The upper electrode of the MTJ cell is coupled to a bit line BL. In some embodiments, the source line SL can be formed by metal layers M1 and M2, and the bit line BL can be formed by a metal layer M3. In certain embodiments, one of more metal wirings is a single device layer, and in other embodiments, one or more metal wirings are two or more device layers.

FIG. **5**A shows a cross sectional views of a MTJ MRAM according to embodiments of the present disclosure. Material, configuration, dimensions and/or processes the same as or similar to the foregoing embodiments described with FIGS. **1**A-**4**C may be employed in the following embodiments, and detailed explanation thereof may be omitted.

As shown in FIG. **5**A, the MTJ cells of an MRAM are disposed over a substrate **201**. In some embodiments, the substrate **201** includes a suitable elemental semiconductor, such as silicon, diamond or germanium; a suitable alloy or compound semiconductor, such as Group-IV compound semiconductors (e.g., silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), GeSn, SiSn, SiGeSn), Group III-V compound semiconductors (e.g., gallium arsenide (GaAs), indium gallium arsenide (InGaAs), indium arsenide (InAs), indium phosphide (InP), indium antimonide (InSb), gallium arsenic phosphide (GaAsP), or gallium indium phosphide (GaInP)), or the like. Further, the substrate **201** may include an epitaxial layer (epi-layer), which may be strained for performance enhancement, and/or may include a silicon-on-insulator (SOI) structure.

Various electronic devices (not shown), such as transistors (e.g., MOS FET), are disposed on the substrate **201**. The MOS FET may include a planar MOS FET, a fin FET and/or a gate-all-around FET. A first interlayer dielectric (ILD) layer **210** is disposed over the substrate **201** to cover the electronic devices. In some embodiments, another ILD layer **205** is disposed between the first ILD layer and the substrate **201**. The first ILD layer **210** may be referred to as an inter-metal dielectric (IMD) layer. The first ILD layer **210** includes one or more dielectric layers, such as silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, a polymer such as polyimide, combinations of these, or the like. In some embodiments, the first ILD layer **210** is formed through a process such as CVD, flowable CVD (FCVD), or a spin-on-glass process, although any acceptable process may be utilized. Subsequently, a planarization process, such as chemical mechanical polishing (CMP) and/or an etch-back process, or the like is performed.

Further, a lower metal wiring **213** is formed by, for example, a damascene process. The lower metal wiring **213** includes one or more layers of conductive material, such as Cu, a Cu alloy, Al or any other suitable conductive materials.

Each of the MTJ cells is disposed over the lower metal wiring **213**, as shown in FIG. **5**A. Although FIG. **5**A shows two MTJ cells, the number of the MTJ cells is not limited to two. In some embodiments, the lower metal wirings **213** are formed at the N-th metal wiring layer, where N is an integer from any of 2 to 8.

As shown in FIG. **5**A, a first insulating layer functioning as an etch stop layer **220** is formed on the first ILD layer **210**. In some embodiments, the first insulating layer **220** includes a material different from the first ILD layer **210** and includes silicon carbide, silicon nitride, aluminum oxide or any other suitable material. The thickness of the first insulating layer **220** is in a range from about 10 nm to about 25 nm in some embodiments. In some embodiments, an additional insulating layer **222** functioning as an etch stop layer is formed on the first insulating layer **220** and includes silicon carbide, silicon nitride, aluminum oxide or any other suitable material, different from the first insulating layer **220**.

A second ILD layer **225** is formed over the second insulating layer **222**. The second ILD layer includes one or more dielectric layers, such as silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, a polymer such as polyimide, combinations of these, or the like. In some embodiments, the material for the first ILD layer **210** and the material for the second ILD layer **225** are the same. In other embodiments, different dielectric materials are used for the first ILD **210** and the second ILD layer **225**.

A via contact **219** is formed in contact with the lower metal wiring **213** and passing through the second ILD layer **225** and the first and second insulating layers **220**, **222** in some embodiments. In some embodiments, the via contact **219** includes a liner or barrier layer **215** and a body layer **217**. The liner layer **215** includes one or more layers of Ti, TiN, Ta or TaN, or other suitable material, and the body layer **217** includes one or more layers of W, Cu, Al, Mo, Co, Pt, Ni, and/or an alloy thereof or other suitable material, in some embodiments.

An MRAM cell includes a bottom electrode **254**, an MTJ film stack **255** and a top electrode **260**, as shown in FIG. **5**A. The bottom electrode **254** and the MTJ film stack **255** correspond to the first electrode **110** and the MTJ film stack **101** of FIG. **1**B. In some embodiments, the top electrode **260** corresponds to the second electrode **155** of FIG. **1**B or the wiring layer My of FIG. **1**A. In some embodiments, the top electrode **260** includes one or more of Ti, TiN, Ta or TaN. In some embodiments, the width of the bottom electrode **254** is smaller than the largest width of the via contact **219**.

The MRAM cell structure has a tapered shape in some embodiments, as shown in FIG. **5**A. The width of the MRAM cell structure at the bottom (the bottom electrode **254**) is greater than the width at the top. The thickness of the bottom electrode **254** is in a range from about 5 nm to about 20 nm in some embodiments. The thickness of the MTJ film stack **255** is in a range from about 15 nm to about 50 nm in some embodiments. In some embodiments, the width of the bottom electrode **254** is greater than the largest width of the MTJ film stack **255**. The difference in the width between the bottom electrode and the largest width of the MTJ film tack **255** is in a range from about 1 nm to about 5 nm depending on the design and/or process requirement.

In some embodiments, a first insulating cover layer **227** as a sidewall spacer layer is formed on opposing side walls of the MRAM cell structure. The first insulating cover layer **227** includes one or more layers of insulating material. In some embodiments, a nitride-based insulating material is used. In certain embodiments, the nitride-based insulating material is a silicon nitride-based insulating material, such as silicon nitride, SiON, SiCN and SiOCN. The thickness T**1** of the first insulating cover layer **227** (a horizontal largest width) is in a range from about 5 nm to about 30 nm in some embodiments, and is in a range from about 10 nm to about 20 nm in other embodiments. As shown in FIG. **5**A, the first insulating cover layer **227** lands on the bottom electrode **254** and is separated from the second ILD layer **225** and/or the via contact **219** by the bottom electrode **254**.

Further, a second insulating cover layer (sidewall spacer) **280** is formed over the first insulating cover layer **227** in some embodiments. The second insulating cover layer **280** includes one or more layers of insulating material different from the first insulating cover layer **227**. In some embodiments, an aluminum-based insulating material is used. In certain embodiments, the aluminum-based insulating material includes aluminum oxide, aluminum nitride, aluminum oxynitride, aluminum carbide and/or aluminum oxycarbide. In some embodiments, the concentrations of Al, O, C and/or N in the thickness direction are not uniform. In certain embodiments, the concentration of Al gradually decreases from the bottom to the top of the second insulating cover layer **280**, while the concentrations of O, C and/or N gradually increase from the bottom to the top of the second insulating cover layer **280**. The thickness T**2** of the second insulating cover layer **280** is smaller than the thickness T**1** of the first insulating cover layer (a horizontal largest width) in some embodiments. The thickness T**2** is in a range from about 1 nm to about 10 nm in some embodiments, and is in a range from about 3 nm to about 5 nm in other embodiments. As shown in FIG. **5**A, the second insulating cover layer **280** covers the sidewalls of the bottom electrode **254** and is in contact with the second ILD layer **225**. The length of the lateral part of the second insulating cover layer **280** on the second ILD layer **225** is in a range from about 1 nm to about 10 nm depending on the design and/or process requirements.

Moreover, a third insulating cover layer (sidewall spacer) **285** is formed over the second insulating cover layer **280**. In some embodiments, the third insulating cover layer **285** includes an oxide-based insulating material. In certain embodiments, the oxide-based insulating material is a silicon oxide-based insulating material, such as silicon oxide, SiON, SiOC and SiOCN. In some embodiments, the third insulating cover layer **285** lands on the second insulating cover layer **280** and is separated from the second ILD layer **225** and/or the via contact **219** by the bottom electrode **254**.

Further, a third ILD layer **230** is disposed in spaces between the MRAM cell structures. The third ILD layer **230** includes one or more dielectric layers, such as silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, a polymer such as polyimide, combinations of these, or the like. In some embodiments, the material for the first ILD layer **210**, the material for the second ILD layer **225** and the material for the third ILD layer **230** are the same. In other embodiments, at least two of them are made of different dielectric materials.

Further, a fourth ILD layer **235** is disposed over the third ILD layer **230**. In some embodiments, the fourth ILD **235** layer is a multiple layer structure and includes a first dielectric layer **232** as an etch stop layer formed on the third ILD layer **230**, a second dielectric layer **234** formed on the first dielectric layer **232**, a third dielectric layer **236** formed on the second dielectric layer **234** and a fourth dielectric layer **238** formed on the third dielectric layer **236**. In other embodiments, the fourth ILD layer is a two or three-layer structure without one or more of the first, second or third dielectric layers.

In some embodiments, the first dielectric layer **232**, the second dielectric layer **234** and the fourth dielectric layer **238** are made of different material than the third dielectric layer **236** and include one or more layers of silicon oxide, silicon nitride, SiON, SiOCN, SiCN, SiC or any other suitable material. In some embodiments, the first dielectric layer **232** and second dielectric layer **234** are made of different materials from each other.

One or more of the first dielectric layer **232**, the second dielectric layer **234** and the fourth dielectric layer **238** include a fluorine-doped silicate glass (FSG), low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, a polymer such as polyimide, combinations of these, or the like.

In some embodiments, the third dielectric layer **236** includes an aluminum-based insulating material, such as, aluminum oxide, aluminum nitride, aluminum oxynitride, aluminum carbide and/or aluminum oxycarbide. In other embodiments, the third dielectric layer includes a Zr or Zn based insulating material (Zr oxide, Zn oxide).

In some embodiments, the material for the first ILD layer **210**, the material for the second ILD layer **225**, the material for the third ILD layer **230** and the material for the third dielectric layer **236** are the same. In other embodiments, at least two of them are made of different dielectric materials. The thickness of the fourth dielectric layer **238** is greater than the thicknesses of the first, second and third dielectric layers in some embodiments.

In some embodiments, an upper electrode **256** passes through the fourth ILD layer **235** and is formed over the top electrode **260**. The upper electrode **256** is made of, for example, Cu, Al, Ta, Ti, Mo, Co, Pt, Ni, W, TiN and/or TaN and/or an alloy thereof or other suitable material. In some embodiments, the upper electrode **256** includes one or more liner or barrier layer **256**A and a body metal layer **256**B. In some embodiments, the liner or barrier layer **256**A is made of Ta, TaN and/or Co, and the body metal layer **256**B is made of Cu or a Cu alloy (e.g., AlCu).

FIG. **5**B shows a cross sectional views of a MTJ MRAM according to embodiments of the present disclosure. Material, configuration, dimensions and/or processes the same as or similar to the foregoing embodiments described with FIGS. **1**A-**5**A may be employed in the following embodiments, and detailed explanation thereof may be omitted.

In some embodiments, the upper electrode **256**C is commonly formed over two or more MRAM cell structures. The materials and/or structures of the upper electrode **256**C are the same as those of the upper electrode **256** of FIG. **5**A. In some embodiments, the common contact **256**C functions as a bit line.

FIGS. **5**C and **5**D show cross sectional views of a MTJ MRAM according to embodiments of the present disclosure. Material, configuration, dimensions and/or processes the same as or similar to the foregoing embodiments described with FIGS. **1**A-**5**B may be employed in the following embodiments, and detailed explanation thereof may be omitted.

In some embodiments, the upper electrode **260** protrudes into one or more layers of the fourth ILD layer **235** and is in contact with the upper electrode **256**.

FIGS. **6**A-**17** show various stages of a sequential manufacturing process of the semiconductor device including an MRAM according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **6**A-**17**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. Material, configuration, dimensions and/or processes the same as or similar to the foregoing embodiments described with FIGS. **1**A-**5**B may be employed in the following embodiments, and detailed explanation thereof may be omitted.

As shown in FIG. **6**A, lower metal wirings **213** are formed in the first ILD layer **210** over the substrate **201**. In some embodiments, via contacts **207** are provided under the lower metal wirings **213**. Then, as shown in FIG. **6**B, a first insulating layer as an etch stop layer **220** is formed over the structure of FIG. **6**A, and a second ILD layer **225** is formed over the first insulating layer **220**.

Further, as shown in FIG. **6**B, via contact openings **223** are formed to expose the upper surface of the lower metal wirings **213**, by using one or more lithography and etching operations.

Subsequently, via contact **219** including layers **215** and **217** are formed, as shown in FIG. **6**C. One or more film forming operations, such as CVD, PVD including sputtering, ALD, electro-chemical plating and/or electro-plating, are performed, and a planarization operation, such as CMP, is performed to fabricate the the via contacts **219**.

In FIGS. **7**-**13**, a MRAM area and a logic circuit area are illustrated. The logic circuit area includes driver circuits, logic function circuit and any other semiconductor circuitry.

As shown in FIG. **7**, a first conductive layer **254**A for the bottom electrode **254** is formed over the structure shown in FIG. **6**C and subsequently, a stacked layer **255**A for the MTJ film stack **255** and a second conductive layer **260**A for a hard mask layer are sequentially formed over the first conductive layer **254**A. In some embodiments, the hard mask layer **260**A includes one or more of Ti, TiN, Ta or TaN. In certain embodiments, the hard mask layer **260**A includes TiN with a thickness in a range from about 30 nm to about 100 nm. In some embodiments, the thickness of the first conductive layer **254**A is in a range from about 1 nm to about 5 nm.

Further, one or more hard mask layers HM1, HM2 and HM3 are formed over the second conductive layer **260**A. In some embodiments the first hard mask layer HM1 includes TEOS or silicon oxide having a thickness of about 20 nm to about 35 nm. In some embodiments, the second and third hard mask layer HM2 and HM3 includes amorphous carbon, amorphous silicon, polysilicon, silicon nitride, SiON, SiOCN, SiOC, SiCN, SiC, aluminum oxide, aluminum nitride, hafnium oxide, zinc oxide, zirconium oxide, titanium oxide, or any other suitable material. In some embodiments, the second hard mask layer HM2 includes amorphous carbon having a thickness of about 25 nm to about 40 nm. In some embodiments, the third hard mask layer HM3 includes amorphous silicon having a thickness of about 8 nm to about 20 nm. Further, a photo resist pattern PR is formed on the third hard mask layer HM3.

By using one or more etching operations, the hard mask layer **260**A is patterned into the hard mask pattern **260** as shown in FIG. **8**. Then, by using the hard mask pattern **260** as an etching mask, the stacked layer **255**A and the first conductive layer **254**A are patterned into MRAM cell structures each including the bottom electrode **254**, the MTJ film stack **255** and the hard mask pattern **260**. As shown in FIG. **9**, the etching of the MTJ film stack layer **255**A stops at the first conductive layer **254**A at both the memory cell region and the logic circuit region. In some embodiments, the etching of the MTJ film stack includes reactive ion etching (RIE) or ion beam etching (IBE) or the combination thereof. In some embodiments, IBE is first applied to etch the MTJ film stack and then the etching is switched to RIE, which has a higher etching selectivity between the MTJ film stack and the first conductive layer **254**A (e.g., TiN) than the IBE. In some embodiments, only RIE is used. By using RIE, it is possible to suppress shadowing effects caused by a high aspect ratio of adjacent cell structures, compared with IBE. Moreover, the present embodiments can suppress the loading effect during the etching between the memory cell region (high pattern density) and the logic circuit region (low pattern density).

Subsequently, as shown in FIG. **9**, an insulating layer **227**A for a first insulating cover layer (sidewall) **227** is formed to cover the MRAM cell structures and over the first conductive layer **245**A. The insulating layer **227**A can be formed by CVD, PVD or ALD or any other suitable film deposition method. In some embodiments, the insulating layer **227**A is formed by CVD, PVD or ALD at a temperature range less than about 150° C., such as a range from about 100° C. to about 150° C. When the insulating layer **227**A is formed at a higher temperature, such as a range from about 200° C. to about 300° C. (or more), the film formation process may cause damage to the MTJ film stack **255** since the insulating layer is directly formed on the MTJ film stack **255**. As shown in FIG. **9**, the insulating layer **227**A is conformally formed over the MRAM cell structures in some embodiments.

In some embodiments, an additional insulating layer **227**B is formed over the insulating layer **227**A as shown in FIG. **10**. In some embodiments, the additional insulating layer **227**B includes silicon oxide, SiON, SiOC or SiOCN. The thickness of the additional insulating layer **227**B is in a range from about 2 nm to about 10 nm in some embodiments.

Then, one or more etching operations are performed to partially remove the insulating layer **227**A to form a first insulating cover layer **227** as sidewall spacers, as shown in FIG. **11**. In some embodiments, anisotropic plasma dry etching is employed. As shown in FIG. **11**, the etching also removes the first conductive layer **254**A and stops on the upper surface of the second ILD layer **225** at both the memory cell region and the logic circuit region. Thus, the bottom electrode **254** is formed. As shown in FIG. **11**, a part of the bottom electrode **254** is disposed under the sidewall spacer **227**.

In some embodiments, during the etching of the first conductive layer **254**A, byproducts of the etching (the material of the first conductive layer) are re-deposited over the first insulating cover layer **227**. Since the first insulating cover layer **227** is made of dielectric material while the re-deposited byproducts are conductive, it is possible to selectively remove the byproducts with a high selectivity. In some embodiments, the byproducts are removed by a cleaning operation using wet etching.

Then, as shown in FIG. **12**, an insulating layer **280**A for a second insulating cover layer **280** is formed to cover the MRAM cell structure. The insulating layer **280**A can be formed by CVD, PVD or ALD or any other suitable film deposition method. As shown in FIG. **12**, the insulating layer **280**A is conformally formed. As set forth above, the insulating layer **280**A for the second insulating cover layer **280** includes an aluminum-based insulating material in some embodiments. The aluminum-based insulating material, such as AlO (Al_2O_3), AlN, AlC, AlOC and AlON, can be formed by the following operations. First, an aluminum layer is formed by, for example, metal-organic CVD (MOCVD) or ALD using tri-methyl-aluminum (TMA). Then, a plasma treatment using NH_3, CO_2 and/or CO gases is performed over the aluminum layer, to convert the aluminum layer into AlO, AlN, AlC, AlOC or AlON. The concentrations of Al, O, C and/or N in the plasma treated aluminum layer are not uniform, in particular, along the vertical direction. The AlON layer may be made of two layers of AlO and AlN. In some embodiments, a thin layer of aluminum having a thickness of less than about 1 nm remains at the bottom of the layer. A chemical oxidation of the aluminum layer using an oxidation solution may be employed. In some embodiments, the AlO, AlOC, AlC, AlN and/or AlON layer can be directly formed by CVD, PVD or ALD or other suitable method by using appropriate source gases. In some embodiments, the insulating layer **280**A is formed by CVD, PVD or ALD at a temperature in a range from about 300° C. to about 450° C. Although lower forming temperature (e.g., less than 300° C.) may be employed, since the first insulating cover layer **227** is formed to cover the MTJ film stack **255**, a higher forming temperature (about 300° C. to about 450° C.) may not damage the MTJ film stack **255**.

Next, as shown in FIG. **12**, a dielectric material layer **285**A is formed to fully cover the insulating layer **280**A. In some embodiments, the dielectric layer **285**A includes silicon oxide and is formed by CVD, PVD or ALD.

Subsequently, one or more planarization operations, such as a CMP operation or an etch-back operation, are performed to reduce the height of the dielectric material layer **285**A, and further, an etch-back operation is performed on the dielectric material layer **285**A and the insulating layer **280**A to expose the hard mask pattern **260** and to form the second insulating cover layer **280** and the third insulating cover layer **285**, as shown in FIG. **13**. As shown in FIG. **13**, the etching stops on the upper surface of the second ILD layer **225**, and the top and a part of the side faces of the hard mask pattern **260** are exposed. In some embodiments, the dielectric material layer **285**A and the insulating layer **280**A are fully removed between adjacent memory cell structures, and at the logic circuit region as shown in FIG. **13**.

Then, a dielectric layer for the third ILD layer **230** is formed as shown in FIG. **14**. In some embodiments, the third ILD layer **230** includes one or more dielectric layers and is also formed at the logic circuit region. In some embodiments, the third ILD layer **230** includes a low-k and/or an extreme low-k dielectric material. One or more planarization operations are performed during and/or after the formation of the third ILD layer **230**.

Then, a fourth ILD layer **235** is formed over the third ILD layer **230**, as shown in FIG. **15**. The dielectric layers of the fourth ILD layer can be formed by CVD, PVD or ALD or other suitable film formation method. In some embodiments, the fourth dielectric layer **238** is formed through a process such as CVD, flowable CVD (FCVD), or a spin-on-glass process, although any acceptable process may be utilized. Subsequently, a planarization process, such as chemical mechanical polishing (CMP) and/or an etch-back process, or the like is performed.

Then, as shown in FIG. **16**, contact openings **242** are formed by using one or more lithography and etching operations. In some embodiments, the etching operation removes a part of the hard mask pattern **260**. In some embodiments, the hard mask pattern **260** remains at the bottom of the contact opening **242** as shown in FIG. **16**. In other embodiments, the hard mask pattern **260** is fully removed and the upper surface (the uppermost layer) of the MTJ film stack **255** is exposed at the bottom of the contact opening **242**.

Subsequently, as shown in FIG. **17**, the contact openings **242** are filled with a conductive material so as to form conductive contacts **256** contacting the exposed upper surface of the MTJ film stack **255**. In some embodiments, the conductive contact **256** includes one or more of liner or barrier layers conformally formed on the inner wall of the contact opening **242** and a body metal layer filling the remaining portion of the contact opening. In some embodiments, the liner or barrier layer is made of Ta, TaN and/or Co, and the body metal layer is made of Cu or a Cu alloy (e.g., AlCu).

In some embodiments, after the third ILD layer **230** is formed and before the fourth ILD layer **235** is formed, one or more metal wiring layers including conductive wiring patterns and via contacts are formed in the logic circuit region.

FIGS. **18-21** show various stages of a sequential manufacturing process of the semiconductor device including an MRAM according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **18-21**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. Material, configuration, dimensions and/or processes the same as or similar to the foregoing embodiments described with FIGS. **1A-17** may be employed in the following embodiments, and detailed explanation thereof may be omitted.

In some embodiments, after the third ILD layer **230** is formed, an etch-back operation is performed to expose an upper part of the hard mask layer **260**, as shown in FIG. **18**. Then, a dielectric layer for the first dielectric layer **232** of the fourth ILD layer **235** is formed over the hard mask pattern **260** and the third ILD layer **230**, and then a CMP operation is performed to at least partially expose the hard mask pattern **260**, as shown in FIG. **19**. In some embodiments, the CMP operation stops when the top of the hard mask pattern **260** is exposed or stops after the top of the hard mask pattern **260** is exposed with additional over etching. Then, the second to fourth dielectric layers of the fourth ILD layer are formed over the first dielectric layer **232** as shown in FIG. **19**.

Then, similar to FIGS. **16** and **17**, contact openings **242** are formed by using one or more lithography and etching operations as shown in FIG. **20**. Subsequently, as shown in FIG. **21**, the contact openings **242** are filled with a conductive material so as to form conductive contacts **256** contacting the exposed upper surface of the MTJ film stack **255**.

In the embodiments of the present disclosure, during the etching of the first conductive layer for the bottom electrode, the MTJ film stack is covered by the first insulating cover layer (sidewall insulating layer). Accordingly, it is possible to efficiently remove the etching byproducts that may be deposited on the sidewalls by using proper etchant. Since the sidewall insulating layer is made of a dielectric material while the byproduct is metallic, there are many choices of the etchant. Further by using RIE for the etching of the MTJ film stack, it is possible to suppress shadowing effects caused by a high aspect ratio of adjacent cell structures, compared with IBE. Moreover, the present embodiments can suppress the loading effect during the etching between the memory cell region (high pattern density) and the logic circuit region (low pattern density).

It will be understood that not all advantages have been necessarily discussed herein, no particular advantage is required for all embodiments or examples, and other embodiments or examples may offer different advantages.

In accordance with one aspect of the present disclosure, in a method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, a first layer made of a conductive material is formed over a substrate. A second layer for a magnetic tunnel junction (MTJ) stack is formed over the first conductive layer. A third layer is formed over the second layer. A first hard mask pattern is formed by patterning the third layer. The MTJ stack is formed by patterning the second layer by an etching operation using the first hard mask pattern as an etching mask. The etching operation stops at the first layer. A sidewall insulating layer is formed over the MTJ stack. After the sidewall insulating layer is formed, a bottom electrode is formed by patterning the first layer to form the MRAM cell including the bottom electrode, the MTJ stack and the first hard mask pattern as an upper electrode. In one or more of the foregoing and following embodiments, the conductive material of the first layer is TiN. In one or more of the foregoing and following embodiments, the first hard mask layer is made of TiN. In one or more of the foregoing and following embodiments, the first layer is formed over a first interlayer dielectric (ILD) layer formed over the substrate, and the sidewall insulating layer is not in contact with the first ILD layer. In one or more of the foregoing and following embodiments, in the MRAM cell, a width of the bottom electrode is greater than a largest width of the MTJ stack. In one or more of the foregoing and following embodiments, the sidewall insulating layer is made of silicon nitride. In one or more of the foregoing and following embodiments, in the MRAM cell, a thickness of the bottom electrode is smaller than a thickness of the upper electrode.

In accordance with another aspect of the present disclosure, in a method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, a first interlayer dielectric (ILD) layer is formed over a substrate. A first layer is formed over the first ILD layer. A second layer for a magnetic tunnel junction (MTJ) stack is formed over the first conductive layer. A third layer is formed over the second layer. A first hard mask pattern is formed by patterning the third layer. The MTJ stack is formed by patterning the second layer by an etching operation using the first hard mask pattern as an etching mask. The etching operation stops at the first layer. A first sidewall insulating layer is formed over the MTJ stack. After the first sidewall insulating layer is formed, the first layer is patterned. A second sidewall insulating layer is formed over the first sidewall insulating layer. A third sidewall insulating layer is formed over the second sidewall insulating layer. A second ILD layer is formed, a third ILD layer is formed, a contact opening is formed in the second and third ILD layers, and a conductive layer is formed in the contact opening. In one or more of the foregoing and following embodiments, the first sidewall insulating layer is made of a nitride-based insulating material, and the second sidewall insulating layer is made of an aluminum-based insulating material different from the nitride-based insulating material. In one or more of the foregoing and following embodiments, the nitride-based insulating material is one or more selected from the group consisting of silicon nitride, SiON and SiOCN. In one or more of the foregoing and following embodiments, the nitride-based insulating material is formed at a temperature in a range from 100° C. to 150° C. In one or more of the foregoing and following embodiments, the aluminum-based insulating material is one or more selected from the group consisting of aluminum oxide, aluminum nitride, aluminum oxynitride, aluminum carbide and aluminum oxycarbide. In one or more of the foregoing and following embodiments, the aluminum-based insulating material is formed at a temperature in a range from 300° C. to 450° C. In one or more of the foregoing and following embodiments, the second sidewall insulating layer is in contact with sidewalls of the patterned first layer. In one or more of the foregoing and following embodiments, the sidewall insulating layer is not in contact with the first ILD layer. In one or more of the foregoing and following embodiments, when the third ILD layer is formed, the second ILD layer is partially recessed to expose a part of the hard mask layer, a first dielectric layer is formed over the second ILD layer and the hard mask layer that is exposed, a planarization operation is performed on the first dielectric layer to expose the hard mask layer, and one or more second dielectric layers are formed over the first dielectric layer and the hard mask layer that is exposed.

In accordance with another aspect of the present application, in a method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, a first conductive layer is formed over a first interlayer dielectric (ILD) layer, a stacked layer for a magnetic tunnel junction (MTJ) stack is formed over the first conductive layer, a hard mask pattern is formed over the stacked layer, the stacked layer is patterned by using the hard mask pattern as an etching mask, so as not to expose the first ILD layer, a first insulating cover layer is formed over the patterned stacked layer, the first conductive layer is formed by using the patterned stacked layer with the first insulating cover layer as an etching mask, thereby forming a cell structure including a bottom electrode formed by the first conductive layer, the magnetic tunnel junction (MTJ) stack and an upper electrode formed by the hard mask pattern, a second insulating cover layer and a third insulating cover layer are formed over the cell structure, a second ILD layer is formed, a contact opening is formed in the second ILD layer, and a conductive layer is formed in the contact opening. In one or more of the foregoing and following embodiments, when the first insulating cover layer is formed, a first layer is formed over the patterned stacked layer, a second layer made of a different material than the first layer is formed, and an etch-back operation is performed to exposed the hard mask pattern. In one or more of the foregoing and following embodiments, when the second insulating cover layer and the third insulating cover layer are formed, a first layer for the second insulating layer is formed over the cell structure, a second layer is formed over the first layer, and an etch-back operation is performed to exposed the first ILD layer and the hard mask pattern. In one or more of the foregoing and following embodiments, the second and third insulating cover layer are discontinuous between adjacent cell structures.

In accordance with another aspect of the present disclosure, a semiconductor device including a magnetic random access memory (MRAM) cell, includes: a magnetic random access memory (MRAM) cell structure disposed over a substrate, the MRAM cell structure including a bottom electrode and a magnetic tunnel junction (MTJ) stack; a first insulating cover layer covering sidewalls of the MTJ stack and the bottom electrode; a second insulating cover layer disposed over the first insulating cover layer; a first dielectric layer formed over the second insulating cover layer; a second dielectric layer formed over the first dielectric layer; and a conductive contact formed in the second dielectric layer. A width of the bottom electrode is greater than a largest width of the MTJ stack. In one or more of the foregoing and following embodiments, the first insulting cover layer is made of a nitride-based insulating material, and the second insulting cover layer is made of an aluminum-based insulating material different from the nitride-based insulating material. In one or more of the foregoing and following embodiments, the nitride-based insulating material is one or more selected from the group consisting of SiN, SiON and SiOCN. In one or more of the foregoing and following embodiments, the aluminum-based insulating material is one or more selected from the group consisting of aluminum oxide, aluminum nitride, aluminum oxynitride, aluminum carbide and aluminum oxycarbide. In one or more of the foregoing and following embodiments, the nitride-based insulating material is made of SiN, and the aluminum-based insulating material is one selected from the group consisting of aluminum oxide, aluminum nitride, and aluminum oxynitride. In one or more of the foregoing and following embodiments, the first insulating cover layer is thicker than the second insulting cover layer. In one or more of the foregoing and following embodiments, the semiconductor device further includes a third insulating cover layer disposed between the second insulating cover layer and the first dielectric layer.

In accordance with another aspect of the present disclosure, a semiconductor device including a magnetic random access memory (MRAM) cell, includes: a lower electrode formed in an first interlayer dielectric (ILD) layer disposed over a substrate; a magnetic random access memory (MRAM) cell structure disposed on the lower electrode, the MRAM cell structure including a bottom electrode and a magnetic tunnel junction (MTJ) stack; a first insulating cover layer covering sidewalls of the MTJ stack and the bottom electrode; a second insulating cover layer disposed over the first insulating cover layer; a dielectric layer disposed over the second insulating cover layer; and a conductive contact formed in the dielectric layer. The first insulating cover layer is not in contact with the first ILD layer. In one or more of the foregoing and following embodiments, the first insulating cover layer is separated from the first ILD layer by the bottom electrode. In one or more of the foregoing and following embodiments, the bottom electrode is made of TiN. In one or more of the foregoing and following embodiments, the semiconductor device further includes a third insulating cover layer disposed between the second insulating cover layer and the dielectric layer. In one or more of the foregoing and following embodiments, the third insulating cover layer is not in contact with the first ILD layer. In one or more of the foregoing and following embodiments, the width of the bottom electrode is greater than a largest width of the lower electrode. In one or more of the foregoing and following embodiments, the MRAM cell structure further includes an upper electrode, and the first insulating cover layer covers a part of a side face of the upper electrode. In one or more of the foregoing and following embodiments, the second insulating cover layer is in contact with a side face of the bottom electrode.

In accordance with another aspect of the present disclosure, a semiconductor device including a magnetic random access memory (MRAM) cell, includes: first and second magnetic random access memory (MRAM) cell structures disposed over a substrate, each of the first and second MRAM cell structures including a bottom electrode, a magnetic tunnel junction (MTJ) stack and an upper electrode; a first insulating cover layer covering sidewalls of each of the first and second MRAM cell structures; a second insulating cover layer disposed over the first insulating cover layer; a bottom dielectric layer filling a space between the first and second MRAM cell structures; and an upper dielectric layer disposed over the bottom dielectric layer. The first insulating cover layer is discontinuous between the first MRAM cell structure and the second MRAM cell structure, and the second insulating cover layer is discontinuous between the first MRAM cell structure and the second MRAM cell structure. In one or more of the foregoing and following embodiments, the semiconductor device further includes a common conductive contact in contact with the upper electrode of the first and second MRAM cell structures. In one or more of the foregoing and following embodiments, the first insulating cover layer is made of silicon nitride, and the second insulating cover layer is made of aluminum oxide. In one or more of the foregoing and following embodiments, the semiconductor device further includes a third insulating cover layer disposed over the second insulating cover layer and discontinuous between the first MRAM cell structure and the second MRAM cell structure. In one or more of the foregoing and following embodiments, the lower dielectric layer is in contact with a bottom dielectric layer disposed below the second insulating cover layer, and the third insulating cover layer is not in contact with the bottom dielectric layer.

The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, the method comprising:

forming a first layer made of a conductive material over a substrate, wherein the conductive material of the first layer is TiN;

forming a second layer for a magnetic tunnel junction (MTJ) stack over the first layer;

forming a third layer over the second layer;

forming a first hard mask pattern by patterning the third layer;

forming the MTJ stack by patterning the second layer by an etching operation using the first hard mask pattern as an etching mask, wherein the etching operation stops at a top surface of the first layer;

forming a sidewall insulating layer over the MTJ stack; and

after the sidewall insulating layer is formed, forming a bottom electrode by patterning the first layer to form the MRAM cell including the bottom electrode, the MTJ stack and the first hard mask pattern as a top electrode, wherein the sidewall insulating layer and the bottom electrode are formed by a same anisotropic plasma dry etching operation, the bottom electrode is a planar electrode having a first uniform thickness across a width of the bottom electrode and the width of the bottom electrode is smaller than a largest width of a via contact formed under the bottom electrode, and the top electrode has a second uniform thickness across a width of the top electrode greater than the first uniform thickness of the bottom electrode.

2. The method of claim **1**, wherein the first hard mask pattern is made of TiN.

3. The method of claim **1**, wherein:

the first layer is formed over a first interlayer dielectric (ILD) layer formed over the substrate, and

the sidewall insulating layer is not in contact with the first ILD layer.

4. The method of claim **1**, wherein in the MRAM cell, the width of the bottom electrode is greater than a largest width of the MTJ stack.

5. The method of claim **1**, wherein the sidewall insulating layer is made of silicon nitride.

6. The method of claim **1**, wherein the sidewall insulating layer includes:

a first sidewall insulating layer formed over the MTJ stack;

a second sidewall insulating layer formed over the first sidewall insulating layer; and

a third sidewall insulating layer formed over the second sidewall insulating layer.

7. The method of claim **6**, wherein:

the first layer is formed over a first interlayer dielectric (ILD) layer formed over the substrate.

8. The method of claim **7**, wherein:

the second sidewall insulating layer is in contact with the first ILD layer.

9. A method of manufacturing a semiconductor device including a magnetic random access memory (MRAM) cell, the method comprising:

forming a first interlayer dielectric (ILD) layer over a substrate;

forming a first layer over the first ILD layer;

forming a second layer for a magnetic tunnel junction (MTJ) stack over the first layer;

forming a third layer over the second layer;

forming a first hard mask pattern by patterning the third layer;

forming the MTJ stack by patterning the second layer by an etching operation using the first hard mask pattern as an etching mask, wherein the etching operation stops at a top surface of the first layer;

forming a first sidewall insulating layer over the MTJ stack;

after the first sidewall insulating layer is formed, patterning the first layer, wherein the patterned first layer is a planar electrode having a first uniform thickness across a width of the patterned first layer, the width of the patterned first layer is smaller than a largest width of a via contact formed under the patterned first layer, and the first hard mask pattern has a second uniform thickness across a width of the first hard mask pattern greater than the first uniform thickness of the patterned first layer;

forming a second sidewall insulating layer over the first sidewall insulating layer;

forming a third sidewall insulating layer over the second sidewall insulating layer;

forming a second ILD layer;

forming a third ILD layer, wherein the forming the third ILD layer comprises:

partially recessing the second ILD layer to expose a part of the first hard mask pattern;

forming a first dielectric layer over the second ILD layer and the first hard mask pattern that is exposed;

performing a planarization operation on the first dielectric layer to expose the first hard mask pattern; and

forming one or more second dielectric layers over the first dielectric layer and the first hard mask pattern that is exposed;

forming a contact opening in the second and third ILD layers to remove a top portion of the first hard mask pattern, wherein a remaining portion of the first hard mask pattern has a rectangular shape in a cross-sectional view; and

forming a conductive layer in the contact opening.

10. The method of claim **9**, wherein:

the first sidewall insulating layer is made of a nitride-based insulating material, and

the second sidewall insulating layer is made of an aluminum-based insulating material different from the nitride-based insulating material.

11. The method of claim **10**, wherein the nitride-based insulating material is one or more selected from the group consisting of silicon nitride, SiON and SiOCN.

12. The method of claim **11**, wherein the nitride-based insulating material is formed at a temperature in a range from 100° C. to 150° C.

13. The method of claim **10**, wherein the aluminum-based insulating material is one or more selected from the group consisting of aluminum oxide, aluminum nitride, aluminum oxynitride, aluminum carbide and aluminum oxycarbide.

14. The method of claim **13**, wherein the aluminum-based insulating material is formed at a temperature in a range from 300° C. to 450° C.

15. The method of claim **9**, wherein the second sidewall insulating layer is in contact with sidewalls of the patterned first layer.

16. The method of claim **9**, wherein the first sidewall insulating layer is not in contact with the first ILD layer.

17. The method of claim **9**, wherein:

the first hard mask pattern is made of TiN, and

the second sidewall insulating layer is in contact with the first ILD layer.

18. The method of claim **9**, wherein:

the third sidewall insulating layer is separated from the first ILD layer by the second sidewall insulating layer.

19. The method of claim **9**, wherein in the MRAM cell, the width of the patterned first layer is greater than a largest width of the MTJ stack.

20. The method of claim **9**, wherein the first sidewall insulating layer is made of silicon nitride.

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